

FORM 2

THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003 COMPLETE SPECIFICATION

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See section 10; rule 13

1. TITLE OF INVENTION

A RADIO-FREQUENCY PERCEPTION-AS-INFRASTRUCTURE
SYSTEM AND METHOD FOR DISTRIBUTED THREE-DIMENSIONAL
10 SPATIAL WORLD MODEL GENERATION

2. APPLICANT(S)

1. **Name:** Krishna Prasad Sidheeshwaran

Nationality: Indian

15 **Address:** Flat 101, Sai Brindhavan Residency, Gajularamaram, UMCC
Lane, Quthbullapur, Hyderabad, 500055, Telangana, India.

3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in
20 which it is to be performed.

4. CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from Indian Provisional Patent Application No. 202641045220 filed on April 8, 2026. The disclosure of the said provisional application is incorporated herein by reference to the extent applicable to subject matter disclosed therein.

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5. FIELD OF THE INVENTION

The present invention relates to radio-frequency sensing, distributed spatial computing, electromagnetic environment reconstruction, machine perception, radio-frequency perception infrastructure, integrated sensing and communications,
10 and artificial-intelligence-assisted physical-world modelling.

More particularly the present invention relates to an infrastructure-based network of spatially distributed radio-frequency perception nodes configured to acquire electromagnetic observations from physical environments, establish temporal, frequency and/or phase relationships between spatially separated nodes, spatially
15 register and process said observations, and generate a machine-readable three-dimensional representation of physical space.

The generated representation is referred to herein as an RF Spatial World Model and provides a physical-world perception layer for artificial-intelligence systems, autonomous systems, robotic systems, security systems, industrial systems,
20 analytical systems, digital-twin systems and software applications.

The invention further relates to dynamic perception infrastructure in which uncertainty associated with a spatial world model is used to control subsequent RF sensing operations, including selection of participating nodes, sensing modes, operating spectral layers, transmission characteristics, spatial apertures and
25 computational resources.

6. BACKGROUND OF THE INVENTION

Artificial-intelligence and software systems increasingly perform tasks requiring continuous awareness of physical environments. Conventional implementations generally obtain physical-world information through sensors located on the same
5 machine, vehicle, robot or device that performs the eventual computational task. Such architectures create a number of system specific and generic limitations.

A machine-mounted sensor has a physical sensing aperture limited by the dimensions, position and orientation of the sensor. A vehicle-mounted sensor, for example, observes the environment primarily from the vehicle's instantaneous
10 location. Multiple vehicles traversing the same physical region therefore repeatedly acquire overlapping environmental information independently.

Conventional camera systems additionally depend upon optical visibility and produce image data that can reveal facial characteristics, appearance and other visually identifying information. Their utility can be degraded by darkness, smoke,
15 fog, precipitation, occlusion and other environmental conditions.

Light Detection and Ranging systems provide highly accurate geometric observations under suitable conditions but generally require substantially unobstructed propagation between the sensing system and the observed surface or object.

20 Radio-frequency sensing provides an alternative physical sensing modality capable of operation without conventional optical image acquisition. However, a conventional RF sensing architecture is frequently built around a single radar or localized antenna aperture and consequently remains constrained by the physical geometry of the local sensing system.

25 Distributed radio-frequency systems have been investigated to increase sensing coverage and to combine observations from multiple sensing devices. However, a distributed collection of sensing devices does not by itself establish a persistent

machine-readable model of physical space. Differences between local clocks, oscillators, propagation paths, node coordinates, antenna characteristics and receiver chains can introduce inconsistencies between geographically separated measurements.

- 5 Furthermore, electromagnetic observations frequently contain direct, reflected, scattered and diffracted signal components. Conventional sensing systems may treat multipath components primarily as interference or clutter. Such treatment discards environmental information that can contain information concerning surfaces, boundaries and physical structures.
- 10 A further challenge arises from the inverse nature of electromagnetic scene reconstruction. A single RF observation can correspond to multiple possible physical configurations. Consequently, a useful reconstruction system requires multiple observations, physical constraints, spatial registration, temporal information and mechanisms for handling uncertainty between competing spatial
- 15 hypotheses.

- A still further challenge occurs when physical environments change with time. Buildings, roads, walls and other structural objects may remain substantially stationary, while humans, vehicles, machinery and temporary objects move through the same spatial region. A useful perception infrastructure therefore benefits from
- 20 maintaining a persistent state of physical space rather than independently reconstructing every observation interval without historical context.

There consequently exists a need for an infrastructure architecture in which RF sensing resources are deployed throughout a physical region and cooperate to generate an evolving, machine-readable spatial representation of that region.

- 25 There further exists a need for such infrastructure to operate as a common perception layer available to multiple independent AI and software systems, thereby reducing the requirement for each consuming system to independently sense and reconstruct the same physical environment.

There further exists a need for the sensing infrastructure itself to dynamically alter its sensing configuration according to uncertainty in the generated spatial representation.

5 **7. OBJECTS OF THE INVENTION**

The principal objects of the invention are to provide:

- (a) A radio-frequency perception infrastructure configured to generate a machine-readable three-dimensional representation of physical space;
- 10 (b) A plurality of geographically distributed RF perception nodes configured to collectively observe a physical region and generate a spatial world model;
- (c) A configurable RF node comprising active transmission and/or reception functionality and a receiving array, including a passive receiving aperture, for acquiring electromagnetic observations from the environment;
- 15 (d) A dual-aperture RF node comprising an active transmission aperture and a receiving aperture separated by electromagnetic isolation and/or decoupling structures;
- (e) An RF node capable of operating in active, passive, bistatic, multistatic, cooperative and hybrid sensing modes;
- 20 (f) A multi-spectral sensing architecture configured to use different electromagnetic frequency ranges according to their respective propagation, coverage, penetration and spatial-resolution characteristics;
- (g) A transmitter-provenance architecture in which individual RF transmission sources within a defined spatial perception domain possess distinguishable signal characteristics that permit received signal components to be
25 associated with corresponding transmitting nodes;
- (h) A distributed synchronization architecture configured to establish temporal, frequency and/or phase relationships between geographically separated RF perception nodes;

- (i) A distributed sensing architecture in which adjacent effective RF perception volumes overlap to provide multiple observations of common spatial regions;
- 5 (j) A distributed aperture architecture configured to exploit the physical separation of synchronized nodes to provide an effective sensing geometry exceeding the physical aperture of individual nodes;
- (k) A spatial processing architecture configured to preserve transmitter identity, receiver identity, temporal information, phase information, spatial coordinates and other RF characteristics associated with individual observations;
- 10 (l) A reconstruction architecture configured to maintain multiple candidate spatial hypotheses and progressively update the hypotheses using subsequent RF observations and physical constraints;
- (m) A physics-informed, learned and/or hybrid reconstruction architecture configured to convert complex-valued RF observations into three-dimensional spatial information;
- 15 (n) An RF Spatial World Model comprising geometry, occupancy, material/scattering attributes, kinematic state, temporal state, confidence, uncertainty and observation provenance;
- 20 (o) A dynamic perception infrastructure configured to use uncertainty in the RF Spatial World Model to dynamically reconfigure RF sensing resources;
- (p) An architecture in which additional RF nodes, spectral layers and/or active/passive sensing modes are selected according to a spatial perception requirement;
- 25 (q) A distributed, regional and/or centralized computing architecture configured to process RF observations and maintain the RF Spatial World Model;
- (r) An interface configured to provide the RF Spatial World Model to independent AI, autonomous, analytical, industrial, security and software applications;

- (s) A privacy-oriented physical-world perception architecture that provides spatial and semantic information without requiring conventional optical imagery as the primary perception representation; and
- (t) A historical reconstruction architecture configured to preserve phase-associated RF observations and spatial metadata and subsequently reconstruct a historical spatial state of a physical region.

8. SUMMARY OF THE INVENTION

The present invention provides a Radio-Frequency Perception-as-Infrastructure system comprising a plurality of spatially distributed RF perception nodes and associated synchronization, data-processing, spatial-reconstruction and application-interface subsystems.

The infrastructure operates as a physical-world perception layer. Rather than requiring every consuming vehicle, robot, application or artificial-intelligence system to independently acquire and reconstruct its surrounding environment, the infrastructure produces an RF-derived machine-readable representation of physical space that can be consumed by multiple downstream systems.

Each RF perception node comprises a plurality of RF antenna elements configured for transmission, reception, or both. The antenna elements are arranged according to an antenna geometry selected according to operating frequency, desired spatial resolution, sensing volume, physical dimensions, computational requirements and deployment conditions.

In an embodiment, each node comprises a dual-aperture architecture including an active emitter array and a receiving array separated by an electromagnetic isolation structure. The receiving array includes a plurality of elements arranged as a uniform, non-uniform, sparse, coprime, variable-geometry or other suitable array.

The nodes operate in active and/or passive sensing modes. In active sensing operation, one or more nodes transmit controlled RF signals and one or more nodes receive electromagnetic signals resulting from reflection, scattering, transmission or diffraction through the physical environment. In passive sensing operation, one or more nodes receive ambient electromagnetic energy without intentionally transmitting a sensing waveform. The network further supports mixed configurations in which some nodes transmit while other nodes operate as receivers.

The infrastructure employs multiple spectral layers according to the physical sensing requirements of the deployment. Lower-frequency bands can provide comparatively broad-area environmental observations, propagation around or through selected obstructions and macroscopic spatial constraints. Intermediate-frequency bands provide intermediate propagation and resolution characteristics. Higher-frequency mmWave and sub-terahertz bands provide finer spatial discrimination within regions having suitable propagation conditions.

The spectral layers are used as complementary observations of a common physical region. A lower-frequency reconstruction establishes a coarse spatial prior or environmental constraint, while a higher-frequency reconstruction refines localized geometry, object characteristics and spatial boundaries.\

For active operation, individual transmitting nodes generate distinguishable RF signal signatures. The signature comprises one or more characteristics including carrier-frequency assignment or offset, phase sequence, code sequence, pilot structure, modulation, waveform characteristics, timing structure, chirp parameters, spreading sequence or another identifiable feature. The signal signature is locally unique or sufficiently distinguishable within a defined RF perception domain.

Signal-signature allocations can therefore be reused by geographically separate perception domains.

A receiving node identifies a signal signature contained in an incoming RF observation and associates the observation with an originating transmitter. The resulting observation retains transmitter provenance and receiver provenance.

- 5 The RF nodes are synchronized according to a distributed timing, frequency and/or phase reference. A physical reference network comprising optical fiber, dedicated wired infrastructure, wireless reference links or other reference transport carries one or more synchronization signals. Local nodes further execute calibration operations using reciprocal observations between neighbouring nodes.

- 10 The distributed network is organized into one or more coherence subdomains. Within a coherence subdomain, the relative time, frequency and phase error between participating nodes remains within a predetermined tolerance suitable for a selected coherent signal-processing operation. Outside a coherence subdomain, reconstructed information from separate coherent subdomains is combined through spatial, temporal, semantic, probabilistic and/or non-coherent fusion.

- 15 The RF perception nodes are deployed at spatial separations between approximately 50 metres and 500 metres for the defined infrastructure operating regime.

Adjacent nodes are deployed with overlapping effective RF perception volumes. An effective RF perception volume represents a spatial region in which a node provides sufficient RF observability for a defined sensing or reconstruction task.

- 20 The overlap between adjacent perception volumes permits the same physical region to be observed through multiple transmitter-receiver geometries and thereby provides redundant spatial evidence, improved localization constraints, continuity between adjacent perception domains and improved ability to resolve ambiguous electromagnetic observations

- 25 The overlap is measured according to an applicable overlap criterion, including volumetric overlap, projected-area overlap, linear effective coverage overlaps, high-confidence sensing overlap or reconstruction-quality overlap. In

representative embodiments, the overlap criterion corresponds to approximately 30% to 60% of a selected effective coverage measure.

The physical displacement between synchronized RF nodes provides distributed spatial baselines. The spatial processing architecture combines observations
5 according to known node positions and calibrated temporal/frequency/phase relationships, thereby forming an effective distributed sensing geometry.

The resulting observations include complex amplitude and phase information and can further include time delay, Doppler information, angle-of-arrival, angle-of-departure, polarization, channel-state information, frequency response, scattering
10 characteristics, transmitter identity, receiver identity and spatial metadata.

The processing architecture forms a Spatiotemporal RF Observation Graph associating transmitter provenance, receiver provenance, physical-space hypotheses, acquisition time, propagation characteristics and cross-node correspondence. A graph relationship is retained between an RF observation and
15 each candidate propagation or physical-state hypothesis supported by that observation. The graph is thereby used as a reconstruction-state representation rather than merely as a record of measured RF parameters.

The reconstruction system does not require a single deterministic interpretation of every RF observation. Instead, it maintains multiple candidate physical-state
20 hypotheses and assigns confidence, probability and/or an uncertainty measure to the hypotheses according to their consistency with the acquired RF observations, transmitter-receiver provenance, node geometry, temporal continuity, electromagnetic propagation models and physical constraints. Each candidate hypothesis may retain one or more associated propagation hypotheses and the
25 transmitter-receiver observation relationships that support or contradict the candidate physical state.

Additional observations update the hypotheses. Upon acquisition of an additional observation from a different transmitter-receiver relationship, spectral layer,

polarization state or other sensing configuration, the reconstruction engine updates the confidence of candidate hypotheses, including by increasing or reducing confidence, generating a new hypothesis, merging hypotheses, dividing a hypothesis into alternatives, or eliminating an inconsistent hypothesis. The reconstruction engine can therefore use observation diversity specifically to discriminate between competing physical explanations of the same electromagnetic observation.

The resulting information is maintained within an RF Spatial World Model.

The RF Spatial World Model includes one or more of geometric occupancy, surface geometry, object extent, material/scattering characteristics, spatial position, orientation, velocity, acceleration, object classification, temporal persistence, confidence, uncertainty and provenance information identifying the RF observations contributing to a reconstructed spatial state. An exemplary correspondence between a physical environment and the resulting RF Spatial World Model is illustrated in **FIG. 7**.

The world model is continuously updated. A dynamic perception controller evaluates the confidence and uncertainty of different spatial regions. A region having adequate confidence is maintained using an appropriate sensing allocation. A region having insufficient confidence generates an additional sensing requirement.

The additional sensing requirement can activate adjacent nodes, enlarge a coherent subdomain, change an operating frequency, transition between passive and active operation, modify a waveform, adjust beamforming, alter polarization, modify transmitter-receiver pairing, increase observation duration or allocate additional computational resources.

The infrastructure therefore forms a closed-loop dynamic perception architecture:

**RF observation → spatial inference → world model →
uncertainty estimation → sensing reconfiguration → additional
RF observation → spatial inference → updated world model**

5 The RF Spatial World Model is exposed through an interface to independent computational systems. A downstream artificial-intelligence system can therefore obtain a machine-readable representation of physical space, including objects, surfaces, materials, motion and confidence information, without independently acquiring the entire physical environment through conventional optical sensing.

10 The infrastructure further maintains historical RF observations in embodiments requiring retrospective spatial reconstruction. Phase-associated observations, node geometry, synchronization state and spatial metadata are stored over a selected time interval. In one embodiment, the stored observations are subsequently reprocessed to reconstruct historical spatial states and to generate computational viewpoints corresponding to positions in a reconstructable region at which no physical RF node
15 was located.

The invention thus provides Perception-as-Infrastructure, in which physical-world perception is supplied as an infrastructure service to computational systems.

9. BRIEF DESCRIPTION OF THE DRAWINGS

20 (1) **FIG. 1** Illustrates an overall RF perception-as-infrastructure architecture comprising geographically distributed RF perception nodes, a synchronization and calibration layer (110), an RF observation acquisition and pre-processing layer (120), a regional and/or central reconstruction engine (130), an RF Spatial World Model (140), a dynamic perception
25 controller (150), and downstream applications (160), with information flowing between the functional layers to acquire, synchronize, process,

reconstruct and maintain a machine-readable representation of a physical environment.

- 5 (2) **FIG. 2** Illustrates geographically distributed RF perception nodes having overlapping effective RF perception volumes and organized into multiple coherence subdomains within a common spatial coordinate framework, wherein participating nodes within each coherence subdomain provide phase-coherent processing and distributed sensing geometry, and spatial information from different coherence subdomains is combined through spatial or non-coherent fusion to update an RF Spatial World Model.
- 10 (3) **FIG. 3** Illustrates transformation of distributed transmitter-receiver RF observations into a Spatiotemporal RF Observation Graph, generation of candidate physical spatial hypotheses and candidate electromagnetic propagation hypotheses, learned inverse reconstruction, electromagnetic forward prediction, residual and consistency evaluation, hypothesis updating, and maintenance of the RF Spatial World Model.
- 15 (4) **FIG. 4** Illustrates a canonical RF-to-World-Model closed loop in which competing physical spatial and/or electromagnetic propagation hypotheses are evaluated under candidate RF interrogation configurations, hypothesis-specific RF responses are predicted, a configuration is selected according to predicted distinguishability, an RF observation is acquired, hypotheses are evaluated and updated, and the resulting RF Spatial World Model state and uncertainty are used by a dynamic perception controller to select a subsequent sensing configuration.
- 20 (5) **FIG. 5** Illustrates a multi-spectral and multi-resolution reconstruction process in which a lower-frequency reconstruction establishes spatial, occupancy, boundary and/or electromagnetic propagation constraints that restrict or otherwise condition a higher-frequency reconstruction, followed by spatial registration and fusion into the RF Spatial World Model.
- 25 (6) **FIG. 6** Illustrates an operational closed-loop RF perception process in which an RF Spatial World Model provides spatial state and uncertainty information to a Dynamic Perception Controller, which selects an RF
- 30

sensing configuration, followed by distributed RF signal acquisition and pre-processing, reconstruction and hypothesis updating, with the resulting state being used to update the RF Spatial World Model and determine subsequent sensing configurations.

- 5 **(7) FIG. 7** illustrates a representative correspondence between a physical environment and a reconstructed RF Spatial World Model, wherein an environment comprising structures, roadways, vehicles, vegetation, pedestrians and other physical entities is represented in a three-dimensional machine-readable spatial form.

10

10. DEFINITIONS

For purposes of the present specification:

(a) RF Perception Node

- 15 An infrastructure device comprising one or more RF transmitters, RF receivers, antenna elements, synchronization components, calibration components, signal-conversion components, local processing components and communication components.

(b) RF Perception Domain

- 20 A geographically bounded group of RF perception nodes participating in a common spatial sensing and reconstruction operation and sharing defined signal-identification, synchronization and spatial-registration parameters.

(c) Coherence Subdomain

- 25 A subset of RF perception nodes within a perception domain whose relative timing, frequency and/or phase error is maintained within a defined tolerance for a selected coherent-processing operation.

(d) Effective RF Perception Volume

A three-dimensional spatial region in which an RF perception node contributes RF observations having sufficient quality for a defined spatial sensing or reconstruction operation.

5 ***(e) RF Spatial World Model***

A machine-readable representation of physical space generated from one or more RF observations and containing one or more of geometry, occupancy, material characteristics, object state, kinematics, temporal state, confidence, uncertainty and observation provenance.

10 ***(f) Spatiotemporal RF Observation***

An RF observation associated with a transmitter, receiver, spatial coordinate, time, frequency, phase and one or more electromagnetic characteristics.

(g) Transmitter Provenance

15 Information associating a received electromagnetic observation with the RF perception node or transmission source from which the observed signal originated.

(h) Dynamic Perception Infrastructure

20 A sensing infrastructure in which sensing resources are dynamically selected or configured in response to confidence and/or uncertainty in a maintained spatial world model.

(i) RF Interrogation Configuration

A controllable set of one or more RF sensing parameters used to obtain an observation or observation set, including one or more participating nodes,

transmitter-receiver relationships, spectral layers, waveform characteristics, beam states, polarization states, relative transmission phase or amplitude, timing relationships, observation duration, antenna-element subsets and/or other controllable RF operating parameters.

5 ***(j) Candidate Physical Spatial Hypothesis***

A candidate representation of a physical state of at least a portion of the environment, including one or more candidate positions, geometries, occupancies, surfaces, objects, material or scattering characteristics, kinematic states and/or spatial relationships.

10 ***(k) Candidate Electromagnetic Propagation Hypothesis***

A candidate explanation of an observed RF response in terms of one or more direct, reflected, scattered, refracted, diffracted or otherwise propagated signal components, path relationships, propagation conditions and/or electromagnetic interaction with the physical environment.

15 ***(l) Reconstructability State***

Information associated with RF observations that preserves one or more parameters required to interpret or reconstruct those observations in their distributed spatial context, including node position, transmitter and receiver identity, timing, frequency, phase, calibration, spatial registration, active sensing configuration and/or candidate physical or propagation hypotheses.

(m) Distinction Criterion

A criterion used to determine whether a candidate RF interrogation configuration is predicted to produce sufficiently different RF responses under two or more candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses, using response separation, likelihood, divergence, residual separation, propagation-path separation, statistical comparison,

information gain, Fisher-information-related measure, classification margin, correlation, rank, conditioning or another selected measure.

11. DETAILED DESCRIPTION OF THE INVENTION

5 A. RF PERCEPTION NODE ARCHITECTURE

Each RF perception node is installed at a substantially known spatial coordinate within an infrastructure deployment.

10 The node comprises an RF front-end and antenna system, digital signal-processing system, synchronization subsystem, calibration subsystem and network interface. The antenna system comprises a plurality of elements. The number of elements is selected according to deployment requirements and includes embodiments having 8, 16, 32, 64, 128, 256, 500 or more elements. The array geometry includes uniform, non-uniform, sparse, coprime, nested, planar, conformal and variable-geometry configurations.

15 The originally disclosed variable-geometry concept provides an embodiment in which different physical element separations support observations at different spatial scales. Such variable-geometry configurations may include relatively smaller inter-element gaps for increased sensitivity to fine spatial variations and relatively larger separations for increased spatial baseline and angular
20 discrimination. The original provisional specifically disclosed multiple antenna-gap values including 5 cm, 15 cm, 20 cm and 25 cm. The present invention is not limited to those dimensions.

The overall architecture of the RF perception infrastructure is illustrated schematically in **FIG. 1**.

25

B. DUAL-APERTURE CONFIGURATION

In one embodiment, each node comprises an active RF transmission aperture; and a passive and/or active RF receiving aperture. The transmission and reception apertures are separated by electromagnetic isolation and/or decoupling structures.

- 5 The isolation structure comprises metallic shielding, absorbing material, metamaterial structures, physical separation, polarization isolation, cancellation circuits or combinations thereof. The active transmission aperture comprises one or more RF power amplifiers and electronically controlled beamforming elements. The receiving aperture comprises multiple antenna elements and
10 associated RF receiver channels. Semiconductor technologies including gallium nitride, silicon-germanium, silicon, indium phosphide and other RF semiconductor implementations are used according to operating frequency and system requirements.

15 C. ACTIVE, PASSIVE AND COOPERATIVE SENSING

- The RF perception infrastructure supports multiple modes. In active mode, an RF perception node transmits a controlled waveform and receives an environmental response. In passive mode, the node receives electromagnetic energy occurring in the environment without intentionally transmitting a sensing waveform. In
20 bistatic mode, a first node transmits and a second spatially separated node receives. In multistatic mode, multiple nodes transmit and/or receive signals associated with a common spatial sensing operation. In hybrid mode, one group of nodes transmits controlled signals while another group operates in passive or cooperative reception. The mode of operation is selected according to regulatory
25 authorization, propagation conditions, energy requirements, required spatial resolution, network loading, privacy requirements and spatial-model uncertainty.

D. MULTI-SPECTRAL PERCEPTION ARCHITECTURE

The infrastructure operates across multiple frequency ranges. A lower-frequency spectral layer provides comparatively broad spatial coverage and can provide macro-scale environmental constraints. An intermediate spectral layer provides intermediate spatial resolution and propagation characteristics. A higher-frequency spectral layer provides finer spatial discrimination and can provide increased sensitivity to small geometric features.

The frequency bands include, by way of example and without limitation:

- i) sub-GHz and low-VHF operating ranges where authorized;
- 10 ii) approximately 3–4 GHz class operating ranges where authorized;
- iii) approximately 5–6 GHz class ranges;
- iv) mmWave ranges including approximately 24–100 GHz; and
- v) sub-terahertz ranges extending to approximately 300 GHz

Particular frequency assignments are selected according to applicable regulatory authorization and deployment conditions. A lower-frequency observation and a higher-frequency observation are spatially registered into a common world-model coordinate system. The reconstruction engine uses the lower-frequency result as a coarse spatial constraint and uses the higher-frequency result to refine spatial boundaries and object characteristics.

20 For example:

$$\mathcal{R}_{macro} = F_{macro}(Y_{low})$$

&

$$\mathcal{R}_{fine} = F_{fine}(Y_{high}, \mathcal{R}_{macro})$$

where Y_{low} represents lower-frequency RF observations and Y_{high} represents higher-frequency RF observations.

The multi-spectral architecture is not limited to two layers.

E. TRANSMITTER-SPECIFIC SIGNAL IDENTIFICATION

5 In controlled-transmission embodiments, each transmitting node is assigned a signal signature within an RF perception domain.

The signature can be generated through carrier-frequency assignment; carrier-frequency offset; phase sequence; code sequence; modulation; pilot structure; chirp characteristic; pulse timing; spreading sequence; subcarrier assignment; waveform identifier; frequency-hopping sequence; and combinations thereof.

10 For example, nodes within one RF perception domain can transmit at closely spaced carrier frequencies such as 2.444 GHz, 2.445 GHz and 2.446 GHz where the selected spacing is supported by the RF synthesizer, receiver bandwidth, phase-noise characteristics, spectral allocation and applicable regulations.

15 The signature allocation is locally scoped. A geographically separated RF perception domain can reuse an otherwise similar signature set. A receiving node identifies the signature of an incoming signal and associates the associated electromagnetic observation with the originating transmitting node. The signal provenance information is carried into the spatial reconstruction pipeline.

20 F. TRANSMITTER-RECEIVER OBSERVATION MODEL

For an active transmission from transmitter i receiver j the received signal is represented by:

$$Y_{ij}(t) = H_{ij}(t)X_i(t) + C_{ij}(t) + N_{ij}(t)$$

where:

- (i) $H_{ij}(t)$ represents an environment-dependent propagation and scattering response;
- (ii) $X_i(t)$ represents the known signal transmitted by node i
- 5 (iii) $C_{ij}(t)$ represents interference and/or additional signal components; and
- (iv) $N_{ij}(t)$ represents measurement noise.

The environment-dependent response $H_{ij}(t)$ contains contributions from physical surfaces and objects through direct, reflected, scattered, refracted and/or diffracted propagation paths.

- 10 The spatial processing engine estimates parameters of $H_{ij}(t)$ from one or more observations.

G. MULTI-PATH AND ENVIRONMENTAL PROPAGATION

- 15 A received RF observation can comprise multiple propagation components. The system therefore represents an observation as follows:

$$Y_{ij}(t) = Y_{direct}(t) + Y_{reflected}(t) + Y_{scattered}(t) + Y_{diffracted}(t) + N(t)$$

- 20 The reconstruction engine evaluates candidate propagation paths rather than assuming that every non-primary component constitutes unusable clutter. Known transmitter provenance, node geometry, temporal synchronization and observed phase relationships provide constraints for evaluating path hypotheses. A candidate path hypothesis is associated with a spatial region and a propagation configuration. The spatial processing engine compares the candidate hypothesis against observations acquired from additional transmitter-receiver combinations. Hypotheses inconsistent with additional observations, physical geometry or applicable propagation constraints are reduced in confidence.
- 25

H. INTER-NODE SYNCHRONIZATION

The infrastructure includes an inter-node timing, frequency and/or phase reference subsystem. As illustrated in **FIG. 1**, the synchronization and calibration layer (110) establishes temporal, frequency and/or phase relationships among participating RF perception nodes. The reference subsystem uses one or more physical or wireless transmission paths.

Physical paths include optical fiber; dark fiber; leased fiber; dedicated copper communication infrastructure; existing telecommunications wiring; dedicated optical wireless links; and other reference-transport infrastructure.

The reference subsystem provides a common or mutually referenced temporal and/or frequency basis. In a coherent subdomain, each node estimates its local oscillator state relative to a common reference as follows:

$$\phi_n(t) = \phi_{ref}(t) + \Delta\phi_{path,n}(t) + \epsilon_n(t)$$

where:

- (i) $\phi_n(t)$ is the local RF oscillator phase;
- (ii) $\phi_{ref}(t)$ is the reference phase;
- (iii) $\Delta\phi_{path,n}(t)$ represents path-dependent variation; and
- (iv) $\epsilon_n(t)$ represents residual phase error.

The node applies a corresponding correction before performing coherent combination.

In embodiments requiring high-coherence distributed processing, the selected temporal synchronization tolerance may be less than one nanosecond, while corresponding frequency and phase tolerances are selected according to

operating frequency, waveform characteristics, spatial baseline, coherent processing requirements and other system conditions.

5 In one embodiment, the reference subsystem provides a common timing reference, a common frequency reference, a phase reference, or a combination thereof. In one embodiment, the reference information is distributed continuously, periodically, on demand, according to a detected synchronization residual, or according to a selected coherent processing operation.

10 In one embodiment, the reference subsystem provides synchronization information to a plurality of geographically distributed nodes, to a selected coherence subdomain, to a subset of participating transmitter and receiver nodes, or to another selected group of RF perception nodes. In one embodiment, the wireless reference path operates as a primary synchronization mechanism, a supplementary calibration mechanism, a periodic recalibration mechanism, or a fallback synchronization mechanism relative to a wired or optical reference.

15 In one implementation, synchronization state associated with an RF perception node comprises one or more of estimated timing offset, frequency offset, phase offset, oscillator state, reference-path condition, calibration state, residual synchronization error, synchronization confidence and/or synchronization validity.

20 The synchronization state is updated using observations from the physical reference path, reciprocal over-the-air calibration, local calibration measurements, historical synchronization state, oscillator characteristics and/or other available reference information. The synchronization subsystem determines a synchronization residual following a reference or calibration
25 operation. Where the estimated residual satisfies a selected tolerance associated with an intended coherent processing operation, the participating nodes may be designated as a coherence-qualified subset.

5 In one embodiment, where the estimated timing, frequency and/or phase residual does not satisfy a selected coherence criterion, one or more corrective operations are performed, including further reference exchange, reciprocal calibration, oscillator correction, delay compensation, phase correction, frequency correction or repetition of the calibration operation. Where the required coherence condition remains unsatisfied, the affected node may be excluded from a selected coherent processing subset while continuing to contribute spatially registered non-coherent or semi-coherent observations to the RF Spatial World Model.

10 In one embodiment, the reference subsystem supports different synchronization regimes within the broader infrastructure. A first group of nodes may operate as a phase-coherent subset for complex-valued field combination, while another group may operate with timing and/or frequency alignment sufficient for spatially registered but non-coherent processing.

15 The infrastructure therefore does not require a single globally phase-locked state extending over the entire deployment. Different coherence subdomains may independently maintain synchronization states and subsequently contribute spatially registered information to a common RF Spatial World Model.

20 I. OPTICAL OR WIRED REFERENCE DISTRIBUTION

In one embodiment, an optical reference signal is transmitted between distributed nodes. The reference signal is received by an optical or electro-optical interface at each node. The local node measures the timing, frequency and/or phase state of the received reference and establishes a local correction relative to the desired synchronization state. A physical reference path can simultaneously carry sensor data and synchronization information or can operate as a dedicated reference channel.

Existing physical communication infrastructure can therefore be reused where suitable.

J. OVER-THE-AIR RECIPROCAL CALIBRATION

5 A local calibration process complements the physical reference transport. A first node transmits a calibration waveform. A second node receives the waveform and determines its timing, frequency and phase relationship. The second node responds using a reciprocal calibration operation.

10 In one implementation, the reciprocal calibration operation comprises a bidirectional reference exchange between a first RF perception node and a second RF perception node. The first node transmits a reference waveform to the second node, and the second node transmits a corresponding reference waveform to the first node. The forward and reverse observations are processed using reciprocity of the propagation path to estimate relative timing offset, frequency offset and/or
15 phase offset between the nodes. Common propagation-delay components are reduced or cancelled to the extent that the forward and reverse propagation conditions are reciprocal over the calibration interval.

Let the first node transmit at time t_1 , the second node receive at t_2 , the second node transmit a corresponding reference at t_3 , and the first node receive at t_4 .
20 Under a reciprocal or substantially symmetric propagation-path assumption, a relative clock-time offset is estimated as:

$$\Delta t_{AB} = \frac{1}{2}[(t_2 - t_1) - (t_4 - t_3)]$$

An estimated common propagation delay is represented as:

$$\tau_{AB} = \frac{1}{2}[(t_2 - t_1) + (t_4 - t_3)]$$

Measured forward and reverse phase differences may be represented by $\Delta\phi_f$ and $\Delta\phi_r$, respectively. After accounting for known or estimated hardware calibration terms h_f and h_r , a relative phase offset is estimated as:

$$\Delta\phi_{AB} = \frac{1}{2} [(\Delta\phi_f - h_f) - (\Delta\phi_r - h_r)]$$

- 5 A relative frequency offset is obtained from the temporal variation of the estimated phase relationship as:

$$\Delta f_{AB}(t) = \frac{1}{2\pi} \frac{d\Delta\phi_{AB}(t)}{dt}$$

- 10 The calibration processor estimates and compensates for node-specific transmit and receive delays, RF hardware phase offsets, oscillator frequency differences, oscillator phase variation and other calibration terms. The resulting synchronization state controls, as applicable, a local oscillator, numerically controlled oscillator, phase-locked loop, frequency synthesizer, digital timing controller or another timing or phase-control element.

- 15 The difference between expected and measured values is used to estimate residual error associated with local oscillator state, RF hardware delay and other node-specific characteristics. The estimated error is applied to subsequent RF observations.

- 20 The calibration operation is repeated periodically, continuously or in response to a detected calibration residual, channel condition, oscillator stability, required coherence tolerance or selected reconstruction operation. A local high-stability oscillator can provide holdover during intervals between calibration exchanges or temporary degradation of the reciprocal reference link.

A residual synchronization error is determined following a calibration exchange. A representative residual phase error is:

$$e_{\phi} = \text{wrap}(\Delta\phi_{AB} - \Delta\phi_{target})$$

For a selected coherent processing operation, the participating node set satisfies:

$$|\Delta t_{AB} - \Delta t_{target}| \leq \epsilon_t, \quad |\Delta f_{AB} - \Delta f_{target}| \leq \epsilon_f, \quad |e_{\phi}| \leq \epsilon_{\phi}$$

5 In embodiments requiring particularly high coherence, the selected temporal tolerance can be less than one nanosecond, although the invention is not limited to a particular numerical synchronization tolerance. A node that does not satisfy the selected coherence criterion can be excluded from the coherent processing subset while continuing to contribute spatially registered non-coherent or semi-coherent observations.

10 In addition to synchronization state, the calibration subsystem maintains a time-varying hardware calibration state associated with each RF perception node. The hardware calibration state represents one or more transmit-chain characteristics, receive-chain characteristics, oscillator characteristics, timing offsets, phase offsets, amplitude responses, frequency responses, propagation delays through
15 known hardware paths, antenna characteristics, mutual-coupling effects and other node-specific RF characteristics.

The hardware calibration state is updated using one or more of reciprocal over-the-air calibration observations, wired or optical reference observations, internally generated calibration signals, loopback measurements, known
20 reference loads, known calibration paths, operating-condition measurements, historical calibration states and observations of persistent reference structures.

In one embodiment, an RF perception node includes an internal calibration path through which a known calibration waveform is transmitted and received without relying on an external physical propagation path. The resulting internal calibration
25 observation provides an estimate of variation attributable to one or more components of the RF transmit chain, receive chain, local oscillator, signal-conversion circuitry, antenna interface, cable path, timing circuitry and associated

hardware, thereby separating hardware-induced variation from variation occurring in the external environment.

The calibration subsystem additionally receives operating-condition measurements associated with the RF perception node. Such measurements include temperature, humidity, supply voltage, operating duration, RF transmit power, oscillator operating state, enclosure conditions and other quantities correlated with variation of RF hardware characteristics. The measurements are supplied to a calibration model, compensation function, learned calibration model, statistical estimator, lookup structure or another calibration-state estimator.

Calibration state is maintained over multiple temporal scales. A relatively fast state represents oscillator phase and frequency variation; an intermediate state represents temperature-dependent transmit and receive phase, amplitude and delay variation; and a slower state represents long-term component, antenna, cable, interconnect or other aging-related characteristics. The invention is not limited to these temporal scales.

A representative measured RF response is therefore represented as:

$$Y_{meas}(t) = C_{rx}(t) H_{env}(x, t) C_{tx}(t) X(t) + N(t)$$

where $C_{tx}(t)$ and $C_{rx}(t)$ represent time-varying transmit-side and receive-side hardware calibration states or operators, $H_{env}(x, t)$ represents an external environment-dependent propagation response, $X(t)$ represents a transmitted, incident or reference signal, and $N(t)$ represents measurement noise and/or other residual components.

The calibration subsystem estimates the hardware calibration state according to:

$$C_{hw}(t) = F_{cal} \left(Y_{loop}(t), Y_{OTA}(t) Z(t) C_{hw}(t - \Delta t) \right)$$

where $\mathbf{Y}_{loop}(\mathbf{t})$ represents one or more internal calibration or loopback observations, $\mathbf{Y}_{OTA}(\mathbf{t})$ represents one or more reciprocal inter-node or over-the-air calibration observations, $\mathbf{Z}(\mathbf{t})$ represents one or more operating-condition measurements, and $\mathbf{C}_{hw}(\mathbf{t} - \Delta\mathbf{t})$ represents a previous hardware calibration state.

5 The function $\mathbf{F}_{cal}$ may comprise a deterministic estimator, adaptive filter, state-space estimator, machine-learning model, statistical model, neural network, lookup structure or combination thereof.

The wireless reference path can operate as a primary synchronization mechanism, a supplementary calibration mechanism, a periodic recalibration mechanism or a

10 fallback mechanism relative to a wired or optical reference. The physical reference and over-the-air calibration therefore operate as complementary synchronization and calibration layers.

K. COHERENCE SUBDOMAINS AND MACRO-LEVEL FUSION

15 An example of distributed coherence subdomains, spatially separated RF perception nodes and fusion between coherence subdomains is illustrated in **FIG. 2**. In the illustrated architecture, coherence subdomains A, B and C are represented by respective processing blocks (210), (220) and (230), each comprising phase-coherent signal processing, distributed aperture synthesis, and

20 local RF observation and feature extraction. Spatial or non-coherent fusion between the coherence subdomains is represented by fusion block (240), which provides spatial registration, feature association, confidence and uncertainty handling, and world-model update. The resulting RF Spatial World Model is represented by block (140).

25 The network is divided into one or more coherence subdomains according to the relative synchronization accuracy required for the intended reconstruction operation. Nodes belonging to a coherence subdomain provide mutually phase-

referenced measurements. The system performs complex-valued field processing using those observations.

A broader infrastructure comprising multiple coherence subdomains uses spatial registration and fusion to combine the resulting spatial information. The broader network therefore does not require one global phase-locked state extending over the entire infrastructure.

This hybrid architecture provides local coherent processing; larger-scale spatial fusion; spatial continuity; distributed perception; and scalable deployment.

10 L. NODE DEPLOYMENT AND OVERLAPPING PERCEPTION VOLUMES

As illustrated in **FIG. 2**, adjacent RF perception nodes are deployed with an inter-node spatial separation within an approximate range of 50 metres to 500 metres. The effective perception volume associated with a node is determined according to propagation conditions, antenna geometry, operating frequency, transmit power, receiver sensitivity, environmental clutter, reconstruction requirements and other system parameters. Adjacent nodes are positioned so that their effective perception volumes overlap.

An overlap metric can be defined by:

$$O_{ij} = \frac{V_i \cap V_j}{\min(V_i, V_j)}$$

for effective perception volumes V_i and V_j or by another appropriate geometric or sensing-quality metric. A representative deployment maintains:

$$0.30 \leq O_{ij} \leq 0.60$$

according to the selected overlap definition.

The overlapping deployment does not require any particular angular relationship between nodes. Instead, the network selects positions that provide multiple useful transmitter-receiver geometries for a common spatial region.

- 5 The overlap provides redundant observations; multiple spatial perspectives; improved localization constraints; increased reconstruction confidence; continuity of tracked objects; cross-validation of spatial hypotheses; and distributed aperture formation.

M. DISTRIBUTED EFFECTIVE APERTURE

- 10 The spatial separation of nodes establishes distributed spatial baselines.

For a subset $\mathcal{S}$ of participating nodes, the effective sensing geometry is determined from:

$$\mathcal{A}_{eff} = F(\{\mathbf{p}_i\}_{i \in \mathcal{S}}, \lambda, \mathbf{G}_{array}, \Phi, \mathcal{O})$$

where:

- 15 (i) $\mathbf{p}_i$ represents node position;
(ii) λ represents operating wavelength;
(iii) $\mathbf{G}_{array}$ represents local antenna geometry
(iv) Φ represents synchronization/calibration state; and
(v) $\mathcal{O}$ represents applicable observation relationships.

- 20 The effective sensing aperture can therefore expand through the inclusion of additional geographically distributed nodes. The infrastructure need not construct one physically continuous antenna spanning the complete deployment region. Instead, distributed RF hardware provides spatially separated measurement locations whose observations are processed as a coordinated

sensing geometry. As illustrated in **FIG. 2**, these distributed spatial baselines provide coordinated spatial measurement geometry for the reconstruction operation.

5 **N. SPATIOTEMPORAL RF OBSERVATION GRAPH**

FIG. 3 illustrates an exemplary processing relationship between distributed transmitter-receiver RF observations, the Spatiotemporal RF Observation Graph, candidate physical spatial hypotheses and candidate electromagnetic propagation hypotheses, and the subsequent spatial reconstruction process.

10 Each processed RF observation is represented with metadata comprising one or more of transmitter identity; receiver identity; time; frequency; phase; spatial position; range; Doppler; angle; polarization; amplitude; waveform identity; propagation hypothesis; and confidence.

15 The observations are organized into a Spatiotemporal RF Observation Graph. The graph associates observations that originate from a common transmitter, terminate at a common receiver, correspond to a common spatial region, occur within a defined temporal interval, represent a candidate common physical object, correspond across different spectral layers and/or share phase or propagation characteristics. The graph provides an intermediate representation between raw
20 RF measurements and the RF Spatial World Model.

A received RF observation is additionally associated with one or more candidate electromagnetic propagation paths. A candidate propagation path comprises a path topology extending from a transmitting node to a receiving node and including zero or more intermediate physical interactions.

The intermediate physical interactions comprise reflection, scattering, diffraction, refraction, transmission, surface interaction, edge interaction, material interaction and/or another electromagnetic interaction with a candidate physical structure.

5 A candidate propagation path is represented by a sequence of spatial interaction locations:

$$\mathcal{P} = \{\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_m\}$$

where $\mathbf{q}_0$ represents a transmitting location, $\mathbf{q}_m$ represents a receiving location and intermediate points represent candidate electromagnetic interaction locations. The corresponding path length is represented as:

10

$$L_p = \sum_{i=1}^m \|\mathbf{q}_i - \mathbf{q}_{i-1}\|$$

The path topology is associated with propagation delay, phase evolution, attenuation, angular relationship, Doppler response, polarization response, material interaction and/or another propagation characteristic.

15 The graph retains different candidate path topologies for the same measured RF observation. A first candidate can represent a direct transmitter-to-receiver path, a second candidate can represent a single-bounce path involving one intermediate surface, and another candidate can represent a higher-order multipath path involving multiple intermediate reflection, scattering, diffraction and/or transmission interactions.

20 Multiple candidate path hypotheses are retained with associated confidence, probability, uncertainty and/or structural state and are progressively resolved as additional RF observations become available. A candidate path is strengthened when its predicted geometry and propagation characteristics remain consistent across multiple observations, nodes, times, spectral layers, polarization states
25 and/or transmitter-receiver relationships.

A candidate path is reduced in confidence when its predicted characteristics are inconsistent with cross-node geometry, temporal evolution, phase behavior, Doppler behavior, spatial registration, material constraints, propagation constraints and/or other physical consistency conditions.

- 5 The graph maintains separate relationships between an RF observation, a candidate physical spatial hypothesis and one or more candidate electromagnetic propagation hypotheses. A measured observation can support a physical hypothesis while simultaneously supporting several alternative propagation explanations.
- 10 A propagation-path hypothesis can be represented as a graph structure having interaction vertices and propagation segments. Each interaction vertex identifies a candidate physical structure or spatial location together with an interaction type, while each propagation segment retains an estimated segment length, propagation delay, phase contribution, attenuation, angular relationship and/or other
- 15 propagation property.

In one embodiment, the graph retains a predicted path relationship even when an intermediate physical interaction has not yet been uniquely assigned to a particular physical-world entity. Subsequent observations can associate an interaction location with a persistent surface, structure or object in the RF Spatial

20 World Model.

Multiple transmitter-receiver observation relationships are used to test whether a candidate propagation path corresponds to an actual physical entity or instead represents a propagation artifact. A path-associated observation that remains spatially, temporally and geometrically consistent with a persistent physical-

25 world entity is associated with that entity, while a transient or geometrically inconsistent apparent location remains represented as a propagation-path hypothesis.

A multipath-associated observation is therefore retained as propagation evidence without requiring the system to create an independent physical-world entity for every apparent location generated by multipath propagation.

5 Where a candidate propagation path remains inconsistent with measured observations, known node geometry, temporal continuity, material constraints or other physical constraints, the graph reduces the confidence, probability, admissibility or structural state associated with the path hypothesis.

10 The graph associates path hypotheses with multiple acquired observations over time so that propagation relationships can be tracked independently from physical entities inferred from those observations. A path relationship can persist even where the corresponding physical-world interpretation changes, and a physical-world entity can remain persistent even where one particular propagation path becomes unavailable.

15 Subsequent observations from another receiving node, another transmitting node, another frequency, another time, another polarization state or another observation relationship can increase, reduce, merge, divide or eliminate a candidate propagation-path hypothesis. The resulting graph state is supplied to the spatial reconstruction engine, which uses graph relationships, candidate path topologies, distributed node geometry, temporal information and applicable physical
20 constraints to generate and update the RF Spatial World Model.

O. RF SIGNAL PROCESSING

As illustrated in **FIG. 1**, the RF observation acquisition and pre-processing layer (120) receives and conditions RF observations prior to spatial reconstruction. The
25 local and distributed processing architecture executes one or more of filtering; denoising; automatic gain processing; synchronization; phase correction; amplitude calibration; mutual-coupling correction; matched filtering; range

processing; Doppler processing; angle-of-arrival estimation; angle-of-departure estimation; beamforming; STAP; polarization processing; channel-state estimation; interference suppression; signal-source identification; and spatial registration.

- 5 For example, an antenna-array observation may be represented as:

$$\mathbf{Y}_{measured} = \mathbf{C}\mathbf{Y}_{physical} + \mathbf{N}$$

where $\mathbf{C}$ represents an array coupling/calibration operator. A calibration process estimates the appropriate inverse or compensation transform before spatial reconstruction.

- 10 In one embodiment, space-time adaptive processing (STAP) forms an output from a received space-time snapshot according to:

$$\mathbf{y}_{STAP}(\mathbf{t}) = \mathbf{w}^H \mathbf{y}(\mathbf{t})$$

- 15 where $\mathbf{y}(\mathbf{t})$ represents a space-time observation vector and $\mathbf{w}$ represents an adaptive processing weight vector selected according to interference and/or clutter covariance statistics to suppress undesired space-time components while preserving a selected response. STAP may be applied as part of local or distributed RF preprocessing before spatial hypothesis generation, or at another processing stage according to the sensing configuration and reconstruction requirement.

20 **P. ENVIRONMENTAL STATE ESTIMATION**

The infrastructure maintains an environmental state representation. A representative state is $\mathbf{H}_t(\mathbf{x})$, where $\mathbf{x}$ represents a spatial coordinate or spatial cell and $\mathbf{t}$ represents time.

- 25 The state is updated using new RF observations. Persistent environmental structures produce temporally stable observations, while objects moving through

the region produce temporally varying observations. The state-estimation architecture separates these components using temporal continuity, Doppler information, phase variation, spatial persistence and learned or statistical models.

5 The environmental state is maintained separately from a node-specific hardware calibration state and from one or more electromagnetic propagation-state hypotheses. The environmental state represents physical-world properties; the hardware calibration state represents characteristics of the sensing infrastructure; and the propagation-state hypotheses represent candidate explanations for the electromagnetic response between transmitters, receivers and physical structures.

10 Persistent structures, surfaces, buildings, infrastructure objects and other spatial entities are represented as persistent world-model entities having associated geometry, location, material characteristics, temporal persistence, kinematic constraints, confidence and uncertainty.

15 When a subsequent RF observation produces an apparent change in the position, geometry, material response or propagation relationship associated with a persistent physical entity, the reconstruction engine evaluates the change against spatial persistence, geometric continuity, object dimensions, known kinematic properties, temporal continuity, observations from other RF perception nodes, cross-frequency agreement, propagation consistency, spatial registration and
20 previously established world-model state.

In one embodiment, the system maintains competing explanations for a detected RF change, including a physical-environment change hypothesis, an RF propagation-change hypothesis, a hardware-calibration-drift hypothesis and/or a combined hypothesis representing multiple simultaneous changes.

25 Where a persistent building or infrastructure structure is established in the RF Spatial World Model and a subsequent observation indicates an apparent displacement or geometric change inconsistent with established spatial, temporal and physical constraints, the reconstruction engine reduces the confidence of the

physical-change hypothesis and increases the confidence of one or more propagation-state or hardware-calibration hypotheses.

5 A calibration-induced explanation is strengthened when observed RF variation correlates with changes in the estimated hardware calibration state, internal calibration residuals, reciprocal calibration residuals, oscillator state, temperature-dependent calibration parameters or other node-specific calibration measurements. A physical-environment explanation is strengthened when the observed change remains consistent across independently calibrated nodes, transmitter-receiver relationships, spectral layers and/or spatially separated
10 observations.

In one embodiment, additional calibration or sensing operations are selected to distinguish between competing explanations, including an internal loopback calibration measurement, a reciprocal over-the-air calibration exchange, observation of a persistent reference structure, another transmitter-receiver
15 relationship, another spectral layer, another polarization state or acquisition by an additional RF perception node.

The system therefore performs joint estimation of physical-world state, propagation state and hardware calibration state rather than assuming that all changes in measured RF characteristics originate from the physical environment.

20 A representative joint state is represented as:

$$\mathbf{X}_t = \{\mathbf{W}_t, \mathbf{C}_{hw,t}, \mathbf{P}_t \mathbf{\Sigma}_t\}$$

where $\mathbf{W}_t$ represents the RF Spatial World Model, $\mathbf{C}_{hw,t}$ represents one or more estimated hardware calibration states, $\mathbf{P}_t$ represents one or more propagation-state hypotheses, and $\mathbf{\Sigma}_t$ represents associated uncertainty.

This architecture allows the persistence of the physical environment to provide a consistency reference for calibration without requiring the physical environment to be assumed completely static or known with certainty.

5 Q. SPATIAL HYPOTHESIS GENERATION

The RF reconstruction engine generates one or more candidate spatial hypotheses. Each hypothesis H_k includes one or more of object location; object extent; surface geometry; material characteristics; kinematic state; propagation interpretation; and confidence.

10 A probability or confidence value is associated with the hypothesis:

$$P(H_k | R_{1:t})$$

where $R_{1:t}$ represents RF observations acquired over time. Subsequent observations update the hypothesis set.

15 Upon acquisition of a new observation, the reconstruction engine updates the hypothesis set by increasing or reducing confidence, generating a new hypothesis, merging hypotheses, dividing a hypothesis into multiple alternatives, and/or eliminating an inconsistent hypothesis.

This architecture enables the RF reconstruction system to manage the non-unique nature of electromagnetic scene inference.

20

R. SPATIAL RECONSTRUCTION ENGINE

The reconstruction engine maps RF observations into a three-dimensional physical representation. As illustrated in **FIG. 1**, the regional and/or central

reconstruction engine (130) processes the RF observations to generate and update candidate physical spatial hypotheses and candidate electromagnetic propagation hypotheses. The engine comprises one or more of electromagnetic forward models; inverse scattering models; physics-informed neural networks; neural fields; occupancy networks; convolutional neural networks; graph neural networks; transformer architectures; probabilistic inference systems; differentiable propagation models; symbolic constraint systems; and combinations thereof.

A forward model generates predicted RF observations from a candidate spatial state, while a learned inverse model estimates or generates candidate spatial states from measured RF observations. The forward and inverse reconstruction relationship illustrated in FIG. 3 provides a representative implementation of this architecture.

In the illustrated implementation, distributed RF observations are represented by block (310), the Spatiotemporal RF Observation Graph by block (320), candidate physical spatial and candidate electromagnetic propagation hypotheses by blocks (330a) and (330b), the learned inverse model by block (340), the electromagnetic forward model by block (350), residual and consistency evaluation by block (360), hypothesis updating by block (370), and the RF Spatial World Model by block (140). Feedback from the world-model state to subsequent sensing is represented by the connection to the Dynamic Perception Controller (150).

The relationship between candidate generation, hypothesis-specific forward prediction, measured-to-predicted comparison and hypothesis updating can be represented as:

$$\hat{Y}(a, H_k) = F_{EM(H_k, a)}$$

$$H_k \sim F_{inv}(Y)$$

$$\mathbf{R}_{k(a)} = \mathbf{d}\left(\mathbf{Y}_{obs}, \hat{\mathbf{Y}}(\mathbf{a}, \mathbf{H}_k)\right)$$

$$\mathbf{H}_k \leftarrow \text{Update}(\mathbf{H}_k, \mathbf{R}_k, \mathcal{C})$$

where $\mathbf{a}$ represents a selected RF interrogation configuration; $\mathbf{F}_{EM}$ represents an electromagnetic or other forward propagation model; $\mathbf{F}_{inv}$ represents a learned, statistical or hybrid inverse reconstruction process that generates candidate hypotheses from RF observations; $\mathbf{y}_{obs}$ represents an acquired RF observation or observation set; $\mathbf{d}$ represents a selected complex-valued, statistical, geometric, propagation-structure or other discrepancy measure; $\mathbf{R}_{k(a)}$ represents the discrepancy associated with candidate hypothesis $\mathbf{H}_k$ under configuration $\mathbf{a}$; and $\mathcal{C}$ represents one or more applicable physical consistency constraints. The notation $\mathbf{H}_k \sim \mathbf{F}_{inv}(\mathbf{Y})$ denotes that one or more candidate hypotheses may be generated, ranked or otherwise proposed by the inverse reconstruction process rather than requiring a single deterministic spatial state.

The reconstruction objective is represented by:

$$\mathcal{L} = \lambda_r \mathcal{L}_{RF} + \lambda_g \mathcal{L}_{geometry} + \lambda_t \mathcal{L}_{temporal} + \lambda_p \mathcal{L}_{physics} + \lambda_s \mathcal{L}_{semantic}$$

where:

- (i) $\mathcal{L}_{RF}$ represents mismatch between measured and predicted RF observations;
- (ii) $\mathcal{L}_{geometry}$ represents geometric inconsistency;
- (iii) $\mathcal{L}_{temporal}$ represents temporal inconsistency;
- (iv) $\mathcal{L}_{physics}$ represents physical-model inconsistency; and
- (v) $\mathcal{L}_{semantic}$ represents semantic inconsistency.

The parameters $\lambda_r, \lambda_g, \lambda_t, \lambda_p, \lambda_s$ are selected according to the reconstruction task.

For propagation-path reconstruction, a received RF observation is represented as comprising a plurality of candidate propagation paths. A candidate path is represented by a topology $\mathcal{P}_p = \{\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_m\}$, where $\mathbf{q}_0$ represents a transmitting location, $\mathbf{q}_m$ represents a receiving location and intermediate elements represent candidate reflection, scattering, diffraction, transmission or other propagation-interaction locations. An estimated path length is represented as:

$$L_p = \sum_{i=1}^m \|\mathbf{q}_i - \mathbf{q}_{i-1}\|$$

The path topology is associated with propagation delay, phase evolution, attenuation, angular relationship, Doppler response, polarization response, material interaction and/or another propagation characteristic.

For a candidate physical spatial hypothesis H , the reconstruction engine generates candidate propagation-path hypotheses and predicts corresponding RF responses for multiple transmitter-receiver relationships. A composite predicted observation is represented as:

$$Y_H = \sum_{p \in \mathcal{P}_H} \alpha_p F_p(H)$$

where $\mathcal{P}_H$ represents a set of candidate propagation paths associated with hypothesis H , α_p represents a complex path coefficient, and $F_p(H)$ represents a path-specific predicted RF response.

A measured-to-predicted residual is therefore represented as:

$$R_H = d(Y_{obs}, Y_H)$$

where Y_{obs} represents a measured RF observation or observation set and d represents a selected complex-valued, statistical, geometric, propagation-structure or other discrepancy measure.

5 A candidate path or physical hypothesis is strengthened when predicted propagation characteristics remain consistent with observations acquired at multiple nodes and/or multiple times, and is reduced in confidence when the predicted characteristics conflict with cross-node geometry, temporal evolution, phase behavior, Doppler behavior, spatial registration or other physical constraints.

10 In one embodiment, the reconstruction engine maintains competing direct-path, single-bounce and higher-order multipath explanations. A unique propagation interpretation is not required at the first observation; path hypotheses are retained with confidence, probability, uncertainty and/or structural state and progressively resolved as additional RF observations become available.

15 In one embodiment, a multipath-associated observation is associated with a persistent physical-world hypothesis when its temporal evolution and multi-node observation relationships remain consistent with a moving or stationary physical entity. A transient or geometrically inconsistent apparent location remains represented as a propagation-path hypothesis rather than as an independent
20 physical-world entity.

In some embodiments, the measured RF response is evaluated jointly with a time-varying hardware calibration state. A representative relationship is represented as:

$$Y_{obs}(t) = C_{rx}(t) H_{env}(H_k, a, t) C_{tx}(t) X(t) + N(t)$$

25 where $C_{tx}(t)$ and $C_{rx}(t)$ represent transmit-side and receive-side hardware calibration states, H_{env} represents the external environment-dependent

propagation response, $\mathbf{X}(t)$ represents an applied or incident RF signal, and $\mathbf{N}(t)$ represents residual noise or unmodelled components.

5 The reconstruction engine evaluates physical-environment, propagation-state and hardware-calibration explanations jointly. A calibration-drift hypothesis is strengthened when residual variation correlates with internal calibration observations, reciprocal calibration observations, operating-condition changes or node-specific calibration state.

10 In one embodiment, a computationally efficient surrogate of an electromagnetic forward model is used for at least a portion of hypothesis evaluation. The surrogate may comprise a learned neural field, neural radiance field, neural reflectance field, physics-informed neural network, graph-based model, transformer model, reduced-order electromagnetic model, differentiable propagation model, or combination thereof. The surrogate produces a predicted RF response for a candidate physical spatial state and associated propagation hypothesis at lower computational cost than a selected higher-fidelity electromagnetic model.

15 A multi-fidelity reconstruction process first evaluates a plurality of candidate hypotheses using the surrogate and subsequently applies a higher-fidelity electromagnetic or propagation model to a selected subset according to uncertainty, predicted distinguishability, residual error, computational availability, reconstruction importance and/or latency requirements.

20 The reconstruction engine is configured to retain calibration state, propagation hypotheses, observation provenance and model residuals with the corresponding reconstructed spatial state. Candidate states are increased, decreased, retained, split, merged or eliminated according to residual error, physical consistency, temporal continuity, cross-node agreement, cross-frequency agreement, calibration consistency and/or associated uncertainty.

S. PHYSICS-INFORMED INFERENCE

A physical model constrains the learned reconstruction. Physical constraints include electromagnetic propagation consistency; geometric consistency; material property constraints; temporal continuity; kinematic constraints; object-
5 dimension constraints; cross-node consistency; and cross-frequency consistency.

In one embodiment, the physics-informed component is implemented as a PINN, differentiable physics layer, constrained optimization layer, differentiable ray-tracing model or another physics-aware computational architecture.

The present invention does not depend on one particular AI model.

10

T. DOMAIN-AWARE NEURO-SYMBOLIC PROCESSING

A domain-aware processing subsystem receives learned spatial representations and applies deterministic or probabilistic domain constraints. The subsystem is configured to evaluate object dimensions; physically plausible motion; material
15 consistency; spatial connectivity; temporal persistence; cross-node correspondence; and electromagnetic propagation consistency.

A neural processing layer generates candidate spatial clusters. A symbolic and/or deterministic layer evaluates whether the candidate clusters satisfy selected physical constraints. This architecture is referred to in one embodiment as a
20 Domain-Aware Neuro-Symbolic Architecture, or DANA.

DANA is an implementation architecture rather than a limitation on the broader invention.

U. MATERIAL AND OBJECT RECONSTRUCTION

The RF Spatial World Model stores material and object properties inferred from RF observations. RF features used for material inference include amplitude; phase; frequency response; attenuation; polarization; scattering response; propagation delay; Doppler response; spatial extent; and temporal persistence.

5

Candidate material classes include biological structures; metal; concrete; reinforced structures; wood; vegetation; glass; water-containing materials; soil; composite materials; vehicles; machinery; and other physical objects.

In one embodiment, the material inference is probabilistic and provides confidence rather than requiring a deterministic label.

10

V. RF SPATIAL WORLD MODEL

The RF Spatial World Model is a persistent machine-readable representation of a physical region. An exemplary correspondence between a physical environment and its reconstructed RF Spatial World Model is illustrated in **FIG. 7**. As illustrated in **FIG. 1**, the RF Spatial World Model (**140**) maintains reconstructed spatial state together with associated confidence, uncertainty, provenance and candidate hypothesis information. The illustrated RF representation is a model-derived spatial representation generated from distributed RF observations and is not limited to a photographic or optical representation.

15

20

A spatial state can be represented as:

$$W(x, t) = \{G, M, O, K, C, U, P, H\}$$

where:

(i) **G** represents geometric state;

(ii) **M** represents material/scattering state;

25

- (iii) $\mathbf{O}$ represents occupancy;
- (iv) $\mathbf{K}$ represents kinematics;
- (v) $\mathbf{C}$ represents confidence;
- (vi) $\mathbf{U}$ represents uncertainty;
- 5 (vii) $\mathbf{P}$ represents observation provenance; and
- (viii) $\mathbf{H}$ represents candidate spatial hypotheses.

The model is represented by one or more of voxel volume; occupancy grid; point representation; mesh; scene graph; neural field; spatial tensor; graph structure; and combinations thereof.

- 10 The RF Spatial World Model represents both static and dynamic physical entities.

W. TEMPORAL STATE MAINTENANCE

- 15 The infrastructure continuously receives new observations. A previous world-model state is represented as $\mathbf{W}_{t-1}$, and a new observation set as $\mathbf{R}_t$, the current state is updated according to:

$$\mathbf{W}_t = F(\mathbf{W}_{t-1}, \mathbf{R}_t, \mathbf{H}_t, \mathbf{U}_t)$$

where $\mathbf{H}_t$ represents environmental and physical constraints; and $\mathbf{U}_t$ represents uncertainty.

- 20 The update process identifies persistent spatial structure; newly appearing objects; removed objects; moving objects; changed surfaces; uncertain regions; and inconsistent observations. The system therefore maintains a continuously evolving representation of the physical environment.

X. DYNAMIC PERCEPTION INFRASTRUCTURE

The infrastructure evaluates confidence and uncertainty over spatial regions. As illustrated in **FIG. 1**, the dynamic perception controller (**150**) uses the uncertainty and hypothesis state of the RF Spatial World Model to control subsequent sensing and computational resources. A spatial uncertainty function can be represented as $U(\mathbf{x}, t)$, when $U(\mathbf{x}, t)$ remains below a selected uncertainty threshold, the existing sensing configuration continues to maintain the associated world-model state. When $U(\mathbf{x}, t)$ exceeds a selected threshold, the infrastructure generates an additional sensing requirement, as illustrated in **FIG. 1**.

- 10 The additional sensing requirement can cause activation of another RF perception node; activation of neighbouring nodes; enlargement of a coherence subdomain; selection of a different spectral layer; transition from passive to active operation; modification of a transmit waveform; modification of a beamforming pattern; change in polarization; extension of observation duration; change in transmitter-receiver pairing; and increase in processing resources.

The infrastructure thereby operates according to:

$$W_t \rightarrow U_t \rightarrow S_{t+1} \rightarrow R_{t+1} \rightarrow W_{t+1}$$

where S_{t+1} represents the selected subsequent sensing configuration.

20 Y. DYNAMIC NODE AND APERTURE SELECTION

The infrastructure stores available RF sensing resources as a set:

$$\mathcal{S} = \{s_1, s_2, \dots, s_n\}$$

Each sensing resource represents a node, transmitter-receiver combination, spectral layer, waveform, polarization configuration or other sensing parameter.

The controller selects a subset $\mathcal{S}^* \subseteq \mathcal{S}$, according to one or more of expected information gain; spatial uncertainty reduction; energy requirements; computation requirements; communication latency; available spectrum; synchronization constraints; and required reconstruction accuracy.

- 5 The selection can therefore preferentially activate a node whose geometry provides additional spatial information rather than simply activating the nearest node.

Z. MULTI-SPECTRAL ADAPTIVE PERCEPTION

- 10 The dynamic controller can select a spectral layer according to the current world-model uncertainty. A low-frequency layer can be selected when broad-area environmental context is uncertain. A higher-frequency layer can be selected when the coarse structure is known but local geometry requires refinement. Multiple layers can be simultaneously selected when spatial hypotheses remain
15 ambiguous. This allows the infrastructure to dynamically change the scale of perception.

AA. COMPUTATIONAL ARCHITECTURE

The processing architecture is divided into one or more of:

- 20 **(i) *Node-level computation***

Node-level processing includes digitization, filtering, calibration, synchronization, signal identification and local feature extraction. Node-level processing is configured to further perform signal conditioning, beamforming, Doppler processing, feature extraction, candidate-state

generation and/or reduction of received RF data into selected reconstruction-conditioned representations.

(ii) *Regional Computation*

5 Regional computation combines observations from multiple nodes and performs spatial registration and intermediate reconstruction. A regional processor can perform joint hypothesis evaluation, spatial registration, graph construction, candidate hypothesis ranking and deployment-time forward-model evaluation for a selected geographic or computational region.

10 **(iii) *Central Computation***

Central computation performs large-scale spatial reconstruction, AI inference, world-model maintenance, model training, historical reconstruction and application services.

15 Computational execution is distributed across node-level, regional and/or centralized processing locations according to reconstruction latency, communication bandwidth, available computation, energy consumption, synchronization state, observation fidelity, spatial uncertainty, computational cost and reconstruction requirements.

20 In one embodiment, processing is migrated between node-level, regional and centralized locations during operation. Node-level computation handles operations for which low latency, local availability or communication reduction is beneficial, while regional and/or centralized computation handles operations requiring observations from multiple spatially separated nodes, larger computational resources or higher-fidelity
25 reconstruction models.

5 In one embodiment, a node transmits reduced reconstruction-conditioned representations to regional processing while retaining or selectively forwarding higher-fidelity RF information for portions of the environment having unresolved spatial or propagation hypotheses. The sensing configuration and computation configuration can therefore be selected jointly

10 The computational architecture supports the learned inverse model, electromagnetic forward model, surrogate forward model, physics-informed processing and world-model update operations described elsewhere in the specification. The distribution of these operations is not required to remain fixed during operation.

AB. SPATIAL REGISTRATION

15 Each RF perception node has a known local coordinate system. A transformation maps local coordinates into a shared spatial coordinate system:

$$\mathbf{x}_{global} = \mathbf{R}_i \mathbf{x}_{local} + \mathbf{T}_i$$

where $\mathbf{R}_i$ represents orientation, $\mathbf{T}_i$ represents translation.

20 The registration parameters are determined by surveying, calibration, inertial measurements, RF measurements, optical measurements used during installation, or combinations thereof. The resulting observations are expressed in a common global spatial coordinate framework.

AC. OBJECT TRACKING AND INTER-NODE CONTINUITY

A moving object traversing overlapping perception volumes is represented by a persistent world-model entity.

The system performs correspondence between observations originating from neighboring nodes based on spatial continuity; temporal continuity; velocity;
5 acceleration; RF scattering characteristics; phase consistency; Doppler; cross-correlation; cross-ambiguity processing; and transmitter-receiver geometry.

A node does not necessarily create an independent new object each time the object enters its sensing region. The shared world model instead maintains a continuous spatial entity when the observations satisfy selected correspondence conditions.

10

AD. HISTORICAL SPATIAL RECONSTRUCTION

RF observations are stored over a selected rolling time interval. The stored information includes complex I/Q information; phase; timestamps; node coordinates; synchronization state; transmitter identity; receiver identity;
15 waveform identity; calibration state; and spatial registration metadata.

In one embodiment, a rolling buffer overwrites older data under normal operation. An event trigger retains a selected temporal window. The preserved RF history is subsequently processed using the known node geometry and synchronization information. The historical reconstruction engine generates spatial occupancy;
20 reconstructed geometry; object trajectories; material/scattering representations; virtual viewpoints; and temporal replay.

A virtual viewpoint is selected within a spatial region that is reconstructable from the preserved distributed observations. The system therefore reconstructs historical physical states without requiring a physical camera or RF receiver to
25 have occupied the selected viewpoint at the time of the event.

AE. DATA INTEGRITY AND AUTHENTICITY

Where evidentiary or controlled historical reconstruction is required, stored RF observations are associated with integrity metadata. In one embodiment, the node includes a secure hardware subsystem such as a hardware security module. A
5 stored RF observation package is cryptographically hashed and signed.

The package contains one or more of time; RF observation data; node identity; transmitter identity; receiver identity; synchronization state; and spatial metadata.

The cryptographic system provides integrity and provenance for the stored observations. Cryptographic integrity is an optional implementation and does not
10 define the fundamental RF perception architecture.

AF. PRIVACY-ORIENTED PERCEPTION

The RF Spatial World Model does not require a conventional optical image representation.

15 Human-associated spatial entities can be represented as location; geometric extent; pose; orientation; velocity; acceleration; object interaction; material/scattering attributes; and confidence.

A downstream application can therefore consume spatial information without receiving a conventional RGB or optical image. The invention does not require an
20 assumption that RF measurements are incapable of revealing biometric characteristics. Instead, privacy is obtained by controlling the sensing modality, representation and information exposed through the application interface.

AG. PERCEPTION-AS-INFRASTRUCTURE APPLICATION INTERFACE

The RF Spatial World Model is made available through a machine interface. As illustrated in **FIG. 1**, the downstream application interface exposes selected information from the RF Spatial World Model (**140**) to downstream applications (**160**). The interface provides selected spatial information including object location, occupancy, geometry, motion, semantic class, material, confidence, uncertainty, spatial relationships and temporal state.

The interface operates through a local API, network service, data stream, event interface, spatial query service, software-development interface and/or another machine-readable protocol.

The infrastructure therefore separates physical sensing from spatial reconstruction and application intelligence. An external application uses the generated physical-world model without independently executing the entire RF reconstruction pipeline.

In an embodiment, an RF perception node functions as a low-latency edge delivery point for a regional and/or centralized RF Spatial World Model. The node maintains a locally accessible subset, projection, cache or predicted state for a spatial region associated with the node and receives reconstructed state updates from regional and/or centralized processing.

The node can disseminate selected low-latency spatial information to downstream systems, including object position, velocity, trajectory, occupancy, geometry, classification, confidence, uncertainty, environmental state and other control-relevant spatial information. The locally accessible representation can comprise a current state, selected spatial subregion, temporal projection, predicted state, filtered state, task-specific representation or compressed representation.

The edge node can locally interpolate, predict, filter, compress or prioritize the received state according to downstream application requirements, spatial relevance, temporal freshness, confidence, uncertainty, communication bandwidth, latency requirements and available computational resources.

- 5 The interface exposes confidence and uncertainty so that a downstream system distinguishes sufficiently resolved spatial information from provisional, predicted or hypothesis-dependent information. The same underlying RF Spatial World Model is therefore available to multiple independent software systems while the underlying sensing and reconstruction infrastructure remains shared.

10

AH. DOWNSTREAM AI AND SOFTWARE APPLICATIONS

The invention supports multiple downstream applications, for example:

- (i) In autonomous mobility, a vehicle-control system receives infrastructure-derived spatial information
- 15 (ii) In security applications, an analytical model evaluates object relationships and activities in a defined physical region
- (iii) In industrial automation, a robotic controller receives a spatial representation of workers, machinery and equipment
- (iv) In logistics, an application obtains the positions and trajectories of physical assets
- 20 (v) In building systems, an application obtains occupancy and spatial-state information
- (vi) In insurance and incident analysis, an authorized system requests historical observations for a defined spatial and temporal region and
- 25 (vii) In simulation and AI training, the RF Spatial World Model provides structured physical-world states for downstream computational systems

The invention is not limited to these applications.

AI. SCALABLE INFRASTRUCTURE DEPLOYMENT

5 The proposed RF infrastructure is deployable in urban roads; highways; campuses; industrial facilities; ports; airports; warehouses; factories; buildings; tunnels; logistics corridors; public spaces; transportation corridors; and critical infrastructure facilities.

10 The same underlying node and spatial-processing architecture support different deployments through changes in node density; antenna element count; frequency; transmission power; sensing mode; computation placement; overlap geometry; and reconstruction resolution.

AJ. EXEMPLARY URBAN DEPLOYMENT

15 A plurality of RF perception nodes is installed along an urban corridor at inter-node separations within the 50–500 metre operating regime. Adjacent nodes are positioned so that their effective perception volumes overlap. A local group of nodes forms one coherence subdomain. Multiple coherence subdomains are combined into a larger spatial fusion domain.

20 The resulting infrastructure produces an RF Spatial World Model extending throughout the deployment. A downstream AI system consumes the corresponding world model through an application interface. The AI system can independently perform navigation, traffic analysis, security analysis, logistics optimization or another application without changing the underlying RF perception infrastructure.

AK. EXEMPLARY LARGE-AREA DEPLOYMENT

A larger deployment comprises a plurality of RF perception domains. Each domain maintains its own locally coordinated transmitter signatures and coherence subdomains. The signature allocations can be reused between
5 geographically separated domains. The broader infrastructure combines spatial world-model information across the domains.

A large physical region can therefore be covered by a chain or grid of perception domains while maintaining the defined 50–500 metre inter-node regime. The scale of the infrastructure deployment is therefore not limited to the spatial separation
10 of two adjacent RF perception nodes.

AL. EXEMPLARY BUILDING DEPLOYMENT

Within an industrial or commercial building, nodes are deployed at appropriate distances within the defined infrastructure regime or, where physical dimensions
15 require it, the node can be configured with a shorter sensing volume while maintaining the architecture of overlapping spatial observations.

The RF system reconstructs room boundaries; machinery; equipment; people; objects; movement; and occupancy. The resulting world model is consumed by an industrial AI application/s.

20

AM. EXEMPLARY SECURITY APPLICATION

A security application obtains a spatial region from the RF Spatial World Model. The application evaluates spatial and kinematic information for predefined events.

For example, the application identifies an object interaction occurring within a defined region and generates an alert. The RF infrastructure supplies the physical-world representation, while the application-specific model determines which events are operationally relevant.

5

AN. EXEMPLARY AUTONOMOUS SYSTEM APPLICATION

An autonomous system receives a spatial world model corresponding to an area ahead of or around the autonomous platform. The system uses geometry; occupancy; object trajectories; confidence; uncertainty; and material
10 characteristics to calculate a task-specific action or trajectory.

The autonomous system does not require that the infrastructure itself execute the final application-control algorithm.

AO. EXEMPLARY HISTORICAL INCIDENT RECONSTRUCTION

15 An authorized user specifies a spatial region; a start time; an end time; and a reconstruction requirement. The historical reconstruction engine retrieves phase-associated RF observation history for the corresponding region and time interval. The engine performs reconstruction using the distributed node geometry and synchronization information. The resulting historical spatial representation can be
20 rendered from one or more computationally selected viewpoints.

AP. TRAINING AND MODEL DEVELOPMENT

Machine-learning models associated with the spatial reconstruction engine are trained using datasets including RF observations and reference spatial states. The

reference state is obtained by one or more of simulated physical environments; CAD environments; surveyed environments; LiDAR; structured-light systems; optical measurements; synthetic electromagnetic propagation environments; and manually generated reference models.

- 5 During deployment, the inference system operates from RF observations and does not require the reference sensing system used during training. A representative development process uses a physical-world simulation environment and an electromagnetic ray-tracing environment to generate paired RF observations and known scene geometry. A training process learns a mapping between RF
- 10 observations and the reference spatial state. The resulting inference engine can subsequently process RF-only observations to reconstruct the physical environment.

AQ. ERROR, CONFIDENCE AND UNCERTAINTY

- 15 The world model stores uncertainty because electromagnetic scene reconstruction depends upon observation geometry, propagation conditions and measurement quality. Confidence is associated with individual spatial cells, objects, surfaces and spatial hypotheses.

- 20 Uncertainty is derived from sensor agreement, model variance, observation count, geometry, synchronization error, signal-to-noise ratio, cross-frequency disagreement, temporal inconsistency, competing spatial hypotheses, calibration residuals, calibration age and uncertainty associated with the hardware calibration state. The dynamic perception controller uses these values to determine where further sensing or calibration is required.

- 25 Calibration confidence is maintained separately from spatial confidence. A calibration state includes an estimated value, uncertainty, age, residual error, validity state and/or confidence measure.

The system triggers recalibration when calibration uncertainty exceeds a selected threshold, internal calibration residuals increase, reciprocal calibration measurements become inconsistent, environmental-condition changes predict increased hardware variation, synchronization residuals increase, or reconstructed
5 physical changes become inconsistent with established world-model constraints.

In one embodiment, calibration confidence depends upon agreement between internal calibration observations, reciprocal inter-node observations, wired or optical reference observations, persistent reference-structure observations and/or operating-condition measurements.

10 Where a node does not satisfy a selected calibration-confidence or synchronization criterion, the node is excluded from coherent processing while continuing to contribute spatially registered non-coherent or semi-coherent observations. Alternatively, the node is down-weighted according to calibration uncertainty or assigned reduced evidentiary weight in spatial reconstruction.

15 The reconstruction engine maintains separate confidence measures for physical-world hypotheses, electromagnetic propagation hypotheses and hardware-calibration hypotheses. Subsequent observations update these confidence measures according to residual error, physical consistency, cross-node agreement, calibration consistency and temporal continuity.

20 The resulting architecture allows the RF Spatial World Model to act as a physical consistency constraint on calibration, while calibration state is simultaneously used to improve confidence in the RF Spatial World Model. Confidence and uncertainty information are retained with corresponding spatial state, propagation hypothesis, calibration state and observation provenance.

25

AR. IMPLEMENTATION VARIATIONS

The invention includes embodiments in which all nodes are actively transmitting; all nodes operate passively; nodes alternate between active and passive operation; only selected nodes transmit; different frequency layers operate simultaneously; different frequency layers operate at different times; coherent processing occurs within local subdomains; wider-scale fusion is non-coherent; processing occurs entirely at the edge; processing occurs centrally; processing is distributed; neural networks are replaced by another learned model; PINNs are used; neural fields are used; graph networks are used; transformer models are used; symbolic reasoning is used; and combinations of the above are used.

10

AS. INDUSTRIAL APPLICABILITY

The invention is industrially applicable to infrastructures providing machine perception of physical environments, including smart-city systems, transportation infrastructure, autonomous systems, robotics, industrial automation, facilities management, security systems, digital-twin systems, spatial analytics, logistics, simulation platforms, critical infrastructure and other AI-enabled systems requiring persistent machine-readable representations of physical space.

15

AT. CANONICAL RF-TO-WORLD-MODEL PROCESSING PIPELINE

The canonical RF-to-World-Model processing loop is illustrated schematically in **FIG. 4**. The dynamic perception controller evaluates updated uncertainty and determines whether additional sensing is required. When additional sensing is required, the controller selects one or more nodes, transmitter-receiver relationships, spectral layers, waveform parameters, polarization states, observation durations or computational resources.

20
25

New RF observations are acquired and processed through the sensing-reconstruction-update loop. A spatial reconstruction engine evaluates a candidate three-dimensional state using measured RF observations and a forward propagation model. A learned inverse model generates candidate geometry, occupancy, object state and/or material state. Physics-informed and/or domain-aware constraints compare the candidate state with the forward-model prediction and with cross-node, cross-frequency and temporal consistency.

The processed observation retains transmitter identity, receiver identity, acquisition time, frequency, phase-related information, node spatial position, calibration state and measurement-quality information. The observations are organized into the Spatiotemporal RF Observation Graph, including relationships for common transmitters, common receivers, common temporal intervals, candidate common spatial regions, cross-frequency correspondence and candidate propagation relationships.

In one embodiment, a computationally efficient surrogate forward model is used to evaluate a plurality of candidate hypotheses, followed by higher-fidelity forward evaluation of a selected subset. The resulting residuals and consistency information update the candidate hypotheses and the RF Spatial World Model.

The RF Spatial World Model retains reconstructed state together with confidence, uncertainty and provenance. Updated uncertainty is supplied to the dynamic perception controller, thereby closing the loop from RF acquisition through reconstruction and world-model update to the selection of subsequent sensing and computation.

AU. WORLD-MODEL STATE, PROVENANCE AND CONFLICT RESOLUTION

In one embodiment, the RF Spatial World Model records uncertainty associated with degraded regions and marks corresponding spatial state as stale, unresolved or hypothesis-dependent. The dynamic perception controller uses such state to select replacement sensing resources or to reduce computational resolution where
5 available evidence is insufficient.

For at least a selected portion of the infrastructure, the RF Spatial World Model retains hardware calibration state and associated calibration provenance together with reconstructed spatial state. Calibration provenance identifies, as applicable, internal calibration observations, reciprocal calibration observations, operating-
10 condition measurements, calibration residuals, synchronization state, calibration validity interval and node-specific calibration parameters supporting a reconstruction state.

When observations produce conflicting spatial states, the spatial reconstruction engine compares transmitter and receiver provenance, synchronization state, calibration state, spatial registration, temporal relationship, signal quality,
15 propagation hypotheses and physical constraints.

Where a persistent physical entity exhibits an apparent change inconsistent with established world-model constraints, the system compares a physical-change hypothesis against an electromagnetic propagation-change hypothesis, a
20 hardware-calibration-drift hypothesis and/or a combined hypothesis. Calibration residuals, internal loopback observations, reciprocal calibration observations, operating-condition measurements and cross-node consistency provide evidence for resolving the competing explanations.

A hardware-calibration explanation is strengthened when substantially similar changes appear only in observations associated with one node, one transmit or
25 receive chain, one aperture, one oscillator or one calibration state, particularly when the changes correlate with calibration residuals or operating-condition changes. A physical-environment explanation is strengthened when the observed

change remains consistent across independently calibrated nodes, transmitter-receiver relationships, spectral layers, spatially separated observations and established physical constraints.

5 When the conflict remains unresolved, the infrastructure requests additional sensing or calibration operations. Such operations include internal loopback measurement, reciprocal over-the-air calibration, observation of a persistent reference structure, another transmitter-receiver relationship, another RF perception node, another spectral layer, another polarization state, another sensing mode or another observation duration.

10 The RF Spatial World Model maintains a joint reconstruction state comprising physical-world state, propagation-state hypotheses, hardware calibration state, confidence, uncertainty and observation provenance. The joint state is updated as subsequent observations provide evidence for or against the competing explanations.

15 In one embodiment, when a node or coherence subdomain becomes unavailable or its calibration confidence becomes insufficient, overlapping perception volumes and alternative transmitter-receiver relationships continue reconstruction where sufficient observations remain. A sensing resource can be treated as unavailable because of node failure, communication loss, synchronization
20 degradation, interference, propagation blockage, hardware fault, insufficient signal quality or insufficient calibration confidence.

The world model thereby preserves the distinction between measured RF evidence, inferred physical state, electromagnetic propagation hypotheses and RF hardware calibration state.

25

AV. GRACEFUL DEGRADATION AND FAILURE HANDLING

When one or more RF perception nodes or coherence subdomains become unavailable, a plurality of remaining coherence subdomains can continue to operate concurrently. Each remaining subdomain can continue to produce spatial information referenced to a common spatial coordinate framework, and the
5 broader infrastructure can fuse available subdomain outputs using spatial registration, temporal association, object correspondence, uncertainty and/or other RF Spatial World Model state information.

In one embodiment, the infrastructure continues reconstruction using available observations from remaining sensing resources while tolerating the loss or
10 degradation of individual nodes or coherence subdomains. Where sufficient observations remain available, spatially overlapping perception volumes and alternative transmitter-receiver relationships can provide redundant evidence for continued reconstruction of a physical region. The infrastructure is not required to terminate reconstruction of the region solely because one sensing resource or
15 coherence subdomain becomes unavailable.

In one embodiment, where a coherence subdomain becomes unavailable, spatially registered information from other available coherence subdomains continues to contribute to the RF Spatial World Model without requiring one globally phase-locked condition across the entire infrastructure. The infrastructure can thereby
20 maintain broader spatial perception through fusion of available subdomain outputs even when coherent processing cannot be maintained across the complete deployment.

The failure or degradation of a sensing resource can result from node failure, communication loss, synchronization degradation, interference, propagation
25 blockage, hardware fault, insufficient signal quality or another condition that prevents the sensing resource from satisfying the requirements of a selected reconstruction operation.

AW. COHERENT PROCESSING WITHIN SUBDOMAINS AND NON-GLOBAL FUSION

Within a coherence subdomain, phase-referenced observations from participating RF perception nodes are combined using complex-valued processing, coherent
5 beamforming, phase-sensitive correlation, interferometric processing or another phase-preserving operation. Outside a coherence subdomain, spatially registered information from separate subdomains can be fused using spatial registration, temporal association, object correspondence, uncertainty and/or other RF Spatial World Model state information without requiring one globally phase-locked
10 condition across the entire infrastructure.

Subsequent observations from another receiving node, another transmitting node, another frequency, another time or another polarization can increase, reduce, merge, divide or eliminate a candidate electromagnetic propagation hypothesis. This process allows multipath observations to contribute spatial evidence when
15 they remain consistent with the distributed measurement geometry, rather than requiring multipath components to be discarded as undifferentiated clutter.

The spatial reconstruction engine uses the transmitter-receiver relationship as a constraint on candidate propagation paths. For a given observation, the engine can maintain a set of candidate electromagnetic propagation hypotheses
20 corresponding to different direct, reflected, scattered, refracted and/or diffracted paths. Each hypothesis can be associated with one or more candidate spatial regions and a corresponding confidence value.

For controlled-transmission embodiments, a received RF observation is associated with transmitter provenance before the observation is used for spatial inference.
25 The receiving node identifies the signal signature, associates the signal with a transmitting node, records the receiving node and time of acquisition, and stores the resulting transmitter-receiver observation relationship together with the measured RF features.

AX. HIERARCHICAL AND MULTI-RESOLUTION SPATIAL RECONSTRUCTION

FIG. 5 illustrates an exemplary multi-spectral and multi-resolution reconstruction process in which a lower-frequency reconstruction provides spatial and propagation constraints that condition a higher-frequency reconstruction.

In the illustrated processing loop, the RF Spatial World Model (140) provides an updated spatial state, confidence and uncertainty information to the dynamic perception controller (150). The dynamic perception controller (150) determines a subsequent RF sensing or reconstruction configuration, represented by block (560), for obtaining additional observations and/or allocating additional sensing or computational resources according to the remaining uncertainty and reconstruction requirements.

In a multi-spectral embodiment, a lower-frequency reconstruction establishes a coarse occupancy, boundary, region-of-interest or propagation constraint, and a higher-frequency reconstruction uses the constraint to initialize, restrict or regularize a fine-scale reconstruction. The higher-frequency reconstruction is consequently performed over a reduced candidate solution space associated with the lower-frequency result. The resulting fine-scale state is spatially registered with the coarse state and can replace, refine or augment the corresponding region of the RF Spatial World Model.

A spatial region requiring refinement is selected from the uncertainty representation of the RF Spatial World Model. The reconstruction engine allocates additional sensing observations and/or computational resources to the selected region while maintaining a lower-resolution representation elsewhere. Spatial resolution is thereby treated as a dynamically allocated sensing and computation resource rather than as a fixed property applied uniformly across the complete infrastructure.

In one embodiment, the RF Spatial World Model is maintained at one or more spatial resolutions. A coarse-resolution representation is maintained over a broad physical region, while higher-resolution representations are selectively generated for subregions having elevated uncertainty, increased object density, detected motion or a specified reconstruction requirement.

AY. MULTI-DIMENSIONAL OBSERVATION REDUNDANCY

The reconstruction engine combines one or more forms of observation redundancy when determining a spatial state. A candidate spatial state is supported when observations associated with different transmitter-receiver relationships, frequencies, times and/or candidate electromagnetic propagation hypotheses are mutually consistent within a common spatial coordinate system. A candidate state is assigned reduced confidence when the observations required to support the state are inconsistent with measured timing, phase, geometry, propagation constraints or temporal continuity.

Spatial redundancy is obtained from geographically separated RF perception nodes and multiple transmitter-receiver relationships observing a common region. Spectral redundancy is obtained by acquiring observations at multiple operating frequencies or spectral layers having different propagation and spatial-resolution characteristics. Temporal redundancy is obtained by acquiring observations over multiple time intervals and using persistence, motion and temporal consistency. Propagation-path redundancy is obtained by retaining compatible direct, reflected, scattered and diffracted components as candidate evidence rather than discarding all non-primary components.

The spatial reconstruction problem can be underdetermined when observations are obtained from a limited number of transmitter-receiver relationships, a limited frequency range, a limited observation interval or a limited set of spatial viewpoints. The infrastructure therefore provides multiple forms of observation

redundancy, including spatial redundancy, spectral redundancy, temporal redundancy and propagation-path redundancy.

AZ. RECONSTRUCTION-CONDITIONED APERTURE GEOMETRY, 5 MODE CONTINUITY AND CROSS-FREQUENCY CONSTRAINTS

In one embodiment, local antenna geometry is treated as part of the spatial measurement operator used by the reconstruction engine rather than solely as a fixed hardware layout. In one embodiment, the geometry is selected during design, commissioning, reconfiguration or operation according to the physical region
10 being reconstructed, the spatial scale of interest, the candidate physical spatial and/or electromagnetic propagation hypotheses maintained by the reconstruction engine, and the transmitter-receiver relationships available in the surrounding infrastructure.

For a transmitter or receiver element located at spatial position $\mathbf{p}_n$ and an incident
15 or outgoing unit direction $\mathbf{u}$ at operating frequency f , an array-response representation may be written as:

$$\mathbf{a}_n(f, \mathbf{u}) = \exp[-j 2 \pi f (\mathbf{p}_n \cdot \mathbf{u})/c]$$

A collection of such response vectors forms an observation operator $\mathbf{A}$ for a selected frequency, polarization and set of transmitter-receiver relationships. The
20 geometry can therefore be selected or configured so that observations corresponding to candidate physical spatial and/or electromagnetic propagation states produce sufficiently distinct response vectors, or so that the resulting observation operator has a selected rank, conditioning or correlation property for the reconstruction task.

25 The geometry-selection criterion is not limited to a particular optimization. In different embodiments it can use response-vector separation, mutual coherence,

rank, condition number, Fisher-information-related measures, predicted hypothesis separation, propagation-path diversity or another reconstruction-quality criterion. The criterion can be evaluated against a selected set of candidate spatial or electromagnetic propagation hypotheses maintained by the RF Spatial
5 World Model.

In a dual-aperture implementation, the transmission aperture and the receiving aperture are calibrated as distinct signal paths. A calibration model can represent transmit-chain response, receive-chain response, mutual coupling and aperture-specific phase or delay terms. A corrected observation can therefore be represented
10 schematically as:

$$\mathbf{y}_c = \mathbf{C}_{rx} [\mathbf{H}_{env} \mathbf{C}_{tx} \mathbf{x} + \mathbf{L} \mathbf{x}_{tx}] + \mathbf{n}_c$$

where $\mathbf{C}_{tx}$ and $\mathbf{C}_{rx}$ represent calibration operators, $\mathbf{H}_{env}$ represents an environment-dependent propagation response, $\mathbf{L}$ represents residual direct coupling or leakage, and $\mathbf{n}_c$ represents residual noise. The calibrated observation
15 is retained as a complex-valued reconstruction input together with node and aperture provenance.

In one embodiment, the RF infrastructure maintains reconstruction continuity when the sensing mode changes. For example, a region may initially be observed using passive reception and subsequently interrogated using controlled active
20 transmission. The passive and active observations can be registered to the same physical coordinate framework and attached to the same candidate physical spatial and/or electromagnetic propagation hypotheses. Similarly, bistatic or multistatic observations can be incorporated into an existing reconstruction state without creating a separate world model for each sensing mode.

25 In a multi-spectral implementation, the output of a lower-frequency reconstruction can be converted into an admissible set or constraint representation for a higher-frequency reconstruction. The constraint representation can include occupancy

5 masks, geometric boundaries, candidate interaction surfaces, propagation regions, candidate path classes or hypothesis weights. A higher-frequency optimization can then operate over a restricted hypothesis space rather than the unrestricted spatial search volume. The lower-frequency state may thereby exclude, down-weight or regularize high-frequency candidate states that are inconsistent with coarse spatial or propagation evidence.

10 The foregoing embodiments permit the infrastructure to use local aperture design, distributed node geometry, sensing-mode continuity and spectral constraints as coordinated parts of a single spatial reconstruction process. The invention does not require any particular sparse-array taxonomy, antenna-element count, numerical optimization algorithm or frequency assignment.

BA. HYPOTHESIS-DISCRIMINATIVE SENSING CONFIGURATION

15 The hypothesis-discriminative dynamic RF interrogation architecture is illustrated schematically in **FIG. 6**.

20 The dynamic perception mechanism described in this section is an embodiment of the RF Spatial World Model and its reconstruction state. The mechanism does not treat an unresolved region merely as a location requiring a higher signal level or additional generic sensing. Instead, the reconstruction engine can retain competing physical-world hypotheses together with one or more candidate electromagnetic propagation explanations associated with the observations supporting those hypotheses. A subsequent sensing operation can therefore be configured to test a particular unresolved relationship between an observation, a propagation explanation and a physical-world state.

25

For a candidate RF interrogation configuration, the infrastructure determines or estimates a predicted RF response under each of two or more retained hypotheses.

The candidate configuration defines one or more participating transmitters and receivers, their spatial relationship, an operating spectral layer, waveform, polarization, beam state, relative transmission phase or amplitude, timing relationship, observation interval, and/or other controllable RF parameters. The predicted response can include measurable field or channel characteristics and/or a predicted propagation-component structure. The controller is configured to select a configuration when the predicted responses provide a criterion for distinguishing competing hypotheses, rather than selecting the configuration solely according to received power, channel quality or generic coverage.

In one implementation, a discriminability value $D(a, H_i, H_j)$ is determined for a candidate interrogation configuration a and candidate hypotheses H_i and H_j . The value is derived from a difference, divergence, likelihood ratio, predicted measurement separation, propagation-path separation, residual separation, classification margin or another comparison between predicted RF responses. The controller can select a configuration satisfying a discriminability criterion and can additionally apply communication, energy, spectrum, synchronization, safety, hardware and computation constraints. The optimization itself is not required to use any particular information-theoretic metric. The predicted-response comparison and configuration-selection operation is illustrated schematically in FIG. 6.

The predicted RF responses can be generated by an electromagnetic forward model, a learned forward model, a surrogate propagation model, a stored propagation representation, or a combination thereof. In embodiments using controlled distributed transmission, the selected configuration can include coordinated transmission by multiple RF perception nodes. Transmission amplitudes, relative phases, timing, polarization, frequency, beam states or participating nodes can thereby be selected to establish a prescribed interrogation condition for which competing hypotheses are predicted to produce different distributed RF responses. The controlled transmission can be implemented using

coherent or non-coherent coordination according to the applicable coherence subdomain and processing operation.

- 5 A candidate interrogation may correspond to an as-yet-unmeasured observation relationship in the Spatiotemporal RF Observation Graph. Such a candidate relationship can be evaluated computationally before acquisition, using the spatial positions and operating states of participating nodes and the current propagation and physical hypotheses. After acquisition, the relationship becomes a measured graph edge associated with the resulting RF observation and with the hypotheses that the observation supports, contradicts or leaves unresolved.
- 10 The acquired response is associated with the selected observation relationship and inserted into the existing Spatiotemporal RF Observation Graph. The reconstruction engine can compare the acquired response with the predicted responses for the competing hypotheses, update the confidence or uncertainty of the associated propagation hypotheses, and subsequently update the candidate
- 15 physical spatial hypotheses. In some embodiments, a candidate propagation relationship is weakened or eliminated when its predicted observation structure is inconsistent with the acquired observation, while another candidate relationship is retained or strengthened. Thus, the dynamic loop can operate on ambiguity in the reconstruction state itself rather than merely on a generic spatial uncertainty
- 20 magnitude.

BB. HYPOTHESIS-CONDITIONED EFFECTIVE APERTURE FORMATION

- 25 In one embodiment, a distributed RF perception infrastructure treats a participating subset of geographically separated RF perception nodes, antenna elements, transmitters, receivers and associated propagation relationships as a computationally selectable effective aperture. The effective aperture is not limited to a fixed physical array and can change as the reconstruction state changes. A

controller can select a subset of spatially separated nodes and/or local antenna elements to establish transmitter-receiver baselines that are predicted to provide useful discrimination between candidate physical spatial hypotheses or candidate electromagnetic propagation hypotheses maintained by the RF Spatial World Model.

For a selected frequency f and propagation direction $\mathbf{u}$, an antenna element located at $\mathbf{p}_n$ can be represented by an array-response term:

$$\mathbf{a}_n(f, \mathbf{u}) = \exp[-j 2 \pi f (\mathbf{p}_n \cdot \mathbf{u})/c]$$

A collection of participating elements and distributed node relationships defines an observation operator $\mathbf{A}(\mathbf{a})$, where $\mathbf{a}$ denotes a selected RF interrogation configuration. Candidate hypotheses $\mathbf{H}_i$ and $\mathbf{H}_j$ may produce response vectors $\mathbf{A}(\mathbf{a}, \mathbf{H}_i)$ and $\mathbf{A}(\mathbf{a}, \mathbf{H}_j)$, respectively. The controller may therefore select an effective aperture according to response separation, correlation, rank, conditioning, predicted propagation-path separation, a Fisher-information-related quantity, or another reconstruction-quality measure.

In a distributed implementation, physical node locations remain fixed while the computationally selected effective aperture may vary over time. A first aperture can be selected to resolve a coarse positional ambiguity, a second aperture can be selected to distinguish two candidate reflection geometries, and another aperture can be selected to distinguish candidate material or scattering states. The participating aperture may comprise a new node, a previously active node, a different antenna-element subset, a changed transmitter-receiver pairing, or a combination thereof. The selected effective aperture and its active configuration are retained with the resulting observations and reconstruction state so that the origin and geometry of each observation remain available for subsequent inference.

The effective-aperture selection can be performed before acquisition by evaluating candidate observation operators under the current hypothesis set. In this manner,

the infrastructure can avoid selecting a geometry merely because it provides increased signal power or nominal coverage when the selected geometry is expected to produce substantially similar responses for the competing hypotheses. The same mechanism can be used in coherent and non-coherent sensing operations, with
5 coherence-qualified node subsets used where phase-sensitive processing is required and additional nodes contributing spatially registered evidence where coherent combination is not available. For each candidate effective distributed sensing aperture, the controller may evaluate the predicted RF response that the same aperture would produce under each competing reconstruction hypothesis before
10 selecting the aperture, thereby distinguishing aperture selection for resolution of reconstruction ambiguity from aperture selection based solely on signal strength, nominal coverage or general sensing performance.

BC. RECONSTRUCTION-CONDITIONED MULTI-SPECTRAL AND 15 MODE PERTURBATION

The RF interrogation configuration can additionally be treated as a controlled perturbation of the electromagnetic observation conditions. Rather than selecting a spectral layer or sensing mode solely because a region has high uncertainty, the controller can select a frequency band, spectral layer, active or passive mode,
20 waveform, polarization, beam state, relative transmission phase or amplitude, or source-receiver arrangement for which the candidate hypotheses are predicted to produce different measurable consequences.

For example, a region may initially be represented using a lower-frequency reconstruction that constrains occupancy, geometry, boundary location or
25 candidate propagation paths. A higher-frequency interrogation can then be selected not merely to increase resolution, but because the higher-frequency responses predicted for two or more surviving hypotheses differ in a manner useful for resolving the ambiguity. Similarly, a passive observation may leave two

propagation explanations unresolved, after which a controlled active transmitter is introduced from a spatially selected node because the additional source geometry is predicted to make the competing explanations distinguishable.

5 A sensing-mode transition can therefore function as an experimental perturbation of the current reconstruction state. Observations obtained before and after the perturbation remain registered to the same spatial coordinates and associated with the same candidate hypotheses. The response transition itself can be evaluated as evidence: a hypothesis predicting a particular change in propagation delay, phase, polarization response, Doppler behavior, amplitude pattern or path structure can
10 be strengthened when the measured transition is consistent with that prediction and reduced in confidence when the transition is inconsistent.

The controller may evaluate a candidate configuration a using a discriminability quantity defined as:

$$D(a, H_i, H_j) = d(\mathbf{yhat}_i(a), \mathbf{yhat}_j(a))$$

15 where $\mathbf{yhat}_i(a)$ and $\mathbf{yhat}_j(a)$ are predicted RF responses under competing hypotheses, and d is a selected response, likelihood, residual, propagation-structure or statistical comparison. A multi-constraint selection may further account for communication cost, energy, spectrum occupancy, synchronization availability, hardware limitations, safety requirements, latency and computation.

20

BD. RECONSTRUCTION-CONDITIONED OBSERVATION FIDELITY AND COMPUTATIONAL ALLOCATION

In some embodiments, the reconstruction controller determines not only which RF resource should acquire an observation but also what information fidelity must
25 be preserved for the observation to resolve the current ambiguity. A candidate hypothesis pair can differ primarily in phase, delay, polarization, Doppler,

propagation-component structure, amplitude, calibration state or another RF dimension.

5 The reconstruction requirement specifies observation fidelity sufficient to preserve the discriminating information. The requirement includes, by way of example, complex-valued I/Q retention, phase precision, temporal sampling rate, frequency resolution, bandwidth, Doppler resolution, polarization components, channel-state information, propagation-component labels, observation duration, dynamic range and/or another RF information characteristic.

10 The requirement is applied before acquisition or during edge preprocessing so that a local node does not discard information required by a later reconstruction stage. Processing and data placement are therefore coupled to the reconstruction state. A node performs local preprocessing when the retained representation is sufficient for the current hypothesis test, whereas complex RF samples or richer propagation information are forwarded to regional or centralized reconstruction when the
15 ambiguity cannot be resolved from reduced data.

The system selects a sensing configuration and a computational configuration jointly according to predicted reconstruction improvement and resource costs. In one embodiment, computational allocation preserves multiple representations of an observation, including a compact detection representation for routine world-
20 model updates and a higher-fidelity representation selectively retained for a spatial region having unresolved hypotheses.

The resulting data-fidelity decision forms part of the reconstructability state together with the node, transmitter, receiver, spectral, waveform, timing, phase, calibration and aperture configuration used to obtain the evidence.

25 In some embodiments, for real-time operation, computational allocation is additionally conditioned on a cycle-time budget represented as:

$$T_{cycle} = T_{acq} + T_{edge} + T_{comm} + T_{eval} + T_{update}$$

with $T_{cycle} \leq T_{budget}$, where T_{acq} represents acquisition time, T_{edge} represents local or edge-processing time, T_{comm} represents communication time, T_{eval} represents hypothesis and forward-model evaluation time, T_{update} represents world-model update time and T_{budget} represents a selected reconstruction latency budget. The controller is configured to reduce computation, retain lower-cost representations, use a surrogate forward model or move processing between node-level, regional and centralized locations when required to satisfy the cycle-time budget while preserving fidelity necessary for the current reconstruction task.

10 **BE. RF PERCEPTION EXPERIENCE MEMORY**

In some embodiments, the infrastructure maintains an experience record for prior reconstruction experiments. An experience record can associate a physical or propagation condition, one or more unresolved hypotheses, an RF interrogation configuration, predicted responses, acquired responses, reconstruction outcome, resource consumption, and a measure of whether the experiment successfully discriminated the hypotheses. The record can additionally identify environmental, geometric, spectral, synchronization and material conditions under which the experience was obtained.

When a later RF perception domain encounters a reconstruction state having sufficiently similar conditions, a retrieval mechanism can identify one or more prior experience records and propose a corresponding sensing configuration, effective aperture, spectral layer, sensing-mode perturbation, observation fidelity or computational allocation. The retrieved experience is not required to be copied without validation; the current reconstruction engine can evaluate the prior configuration under the current hypotheses and accept, modify or reject it according to predicted discriminability, applicability and confidence.

The experience mechanism therefore permits the distributed infrastructure to accumulate operational knowledge about which RF experiments resolve particular classes of ambiguity. In one example, a first geographic domain can learn that a particular combination of source geometry, material condition and multipath environment makes a candidate propagation explanation difficult to distinguish using one observation relationship but distinguishable using another relationship and polarization. A later domain can use the stored experience as a starting point when a sufficiently similar reconstruction condition occurs, while retaining provenance and confidence regarding the origin and applicability of the prior experience.

Experience records can be maintained separately from deterministic reconstruction outputs and can themselves carry confidence, applicability ranges and outcome statistics. In this manner, the RF infrastructure can learn from prior sensing actions while preserving the distinction between measured evidence, inferred physical state, propagation hypotheses and historical sensing experience.

BF. PRESERVATION OF HISTORICAL INFRASTRUCTURE STATE FOR RECONSTRUCTION

In one embodiment, a computational viewpoint is selected within a region for which the distributed historical observations provide sufficient spatial constraints. The viewpoint is a reconstruction coordinate or virtual sensing position and does not require a physical sensing node to have occupied that coordinate at the historical time.

Accordingly, the historical record is not limited to a prior target track or a fixed image generated at the time of acquisition. In one embodiment, where the preserved RF evidence supports multiple physical interpretations, the historical reconstruction retains or regenerates those alternatives and applies a later reconstruction process to determine whether additional spatial, material,

kinematic or propagation interpretations are supported by the same historical evidence.

5 In one embodiment, the historical record further preserves one or more candidate propagation hypotheses and candidate physical spatial hypotheses generated during or after the observation interval, together with associated confidence and uncertainty. The historical reconstruction engine can use the preserved RF observations and preserved reconstructability state to regenerate a historical world-model state using a reconstruction model that differs from the model used during the original real-time inference.

10 For retrospective reconstruction, the infrastructure can preserve not only RF samples but also a reconstructability state associated with the measurement interval and required to interpret the samples in their original distributed spatial context. The reconstructability state comprises, in various embodiments, node position and orientation, transmitter and receiver identities, signal or waveform
15 identity, timing state, frequency-reference state, phase-reference state, calibration state, spatial-registration state, active transmitter-receiver relationships, sensing configuration, and one or more candidate propagation or physical-state hypotheses associated with the observations.

20 This architecture permits learned inference and electromagnetic modelling to perform complementary functions: the learned inverse component provides candidate-state generation or refinement, while the forward model and physical constraints provide hypothesis-specific predictive consistency. The resulting candidate-state set is maintained as part of the RF Spatial World Model so that subsequent observations can continue the reconstruction process without
25 requiring each observation interval to begin from an independent deterministic reconstruction.

In some embodiments, the learned inverse component proposes a plurality of candidate states, the forward electromagnetic model acts as a consistency

validator of the proposed states, and physical-domain constraints further modify the admissibility of the candidates. A candidate whose predicted RF response is inconsistent with measured observations, transmitter-receiver geometry, timing, propagation behavior, material constraints or temporal continuity can be reduced in confidence or removed, while an alternative candidate can be retained or strengthened.

A reconstruction residual can therefore be represented generally as $\mathbf{R}_i = \mathbf{d}(\mathbf{y}, \mathbf{\hat{y}}_i)$, where, $\mathbf{y}$ is a measured RF observation or observation set, $\mathbf{\hat{y}}_i$ is a forward-model prediction associated with candidate hypothesis $\mathbf{H}_i$, and $\mathbf{d}$ is a selected complex-valued, statistical, geometric, propagation-structure or other discrepancy measure. The residual can be used to modify the confidence, probability, admissibility or structural state of $\mathbf{H}_i$. The residual need not be used solely as a training loss and can be evaluated during deployment-time reconstruction.

For each retained candidate physical spatial state, an electromagnetic forward model generates a predicted RF response for one or more transmitter-receiver observation relationships. The predicted response can include complex-valued amplitude and phase, propagation delay, Doppler, angle, polarization, channel response, spatial field values and/or predicted propagation-component structure. A measured-to-predicted discrepancy is evaluated for individual observations, groups of observations or propagation components.

In an embodiment of the spatial reconstruction engine, the learned inverse component is used to generate candidate physical spatial states from measured RF observations rather than being required to produce a single deterministic scene. A candidate physical spatial state includes geometry, occupancy, material or scattering characteristics, object state, kinematics and one or more candidate electromagnetic propagation explanations. The candidate state is retained together with the RF observations, node geometry and transmitter-receiver provenance that support or contradict the state.

CLAIMS

I claim:

1. A radio-frequency perception infrastructure system for generating and maintaining a machine-readable three-dimensional representation of a physical environment, comprising: a plurality of geographically distributed radio-frequency perception nodes, each comprising one or more radio-frequency transmitters and/or receivers and a plurality of antenna elements, the nodes having known spatial positions within a common spatial coordinate framework; an inter-node synchronization and calibration subsystem configured to establish temporal, frequency and/or phase relationships between at least a subset of the geographically distributed radio-frequency perception nodes; a signal-identification subsystem configured to associate at least a subset of received radio-frequency observations with respective transmitting nodes; a spatial processing subsystem configured to associate respective radio-frequency observations with at least a transmitting node, a receiving node, an acquisition time and a spatial relationship between the transmitting and receiving nodes, and to organize the radio-frequency observations as a spatiotemporal observation structure including relationships between observations and candidate electromagnetic propagation hypotheses and/or candidate physical spatial hypotheses; a spatial reconstruction engine configured to evaluate the spatiotemporal observation structure using multiple transmitter-receiver observation relationships and candidate electromagnetic propagation paths, to maintain a plurality of candidate physical spatial hypotheses together with associated confidence and/or uncertainty information and at least one candidate electromagnetic propagation hypothesis associated with one or more of the candidate physical spatial hypotheses, and to update the candidate propagation and physical spatial hypotheses in response to subsequently acquired radio-frequency observations; and an RF Spatial World Model configured to maintain, for at least a portion of the physical environment, reconstructed spatial state together with at least one of

observation provenance, candidate electromagnetic propagation hypotheses, candidate physical spatial hypotheses, confidence and uncertainty, the RF Spatial World Model being continuously updateable from the radio-frequency observations.

- 5 2. The system as claimed in claim 1, wherein the spatial processing subsystem retains transmitter provenance and receiver provenance identifying transmitting and receiving nodes associated with radio-frequency observations supporting, contradicting or otherwise informing at least a subset of the candidate physical spatial hypotheses and candidate electromagnetic
- 10 propagation hypotheses, and wherein the spatial reconstruction engine uses the retained provenance in evaluating and updating the candidate hypotheses.
- 15 3. A dynamic radio-frequency perception infrastructure system comprising: a plurality of geographically distributed radio-frequency perception nodes configured to acquire radio-frequency observations of a physical environment; a spatial reconstruction engine configured to maintain an RF Spatial World Model comprising a spatial uncertainty representation, a plurality of candidate physical spatial hypotheses for at least one region of the physical environment, and one or more candidate electromagnetic propagation hypotheses associated with the candidate physical spatial hypotheses; and a dynamic perception
- 20 controller configured to identify a region for which two or more candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses remain unresolved, determine a candidate RF interrogation configuration for the region, determine or estimate respective RF responses that would result from the candidate RF interrogation configuration under the unresolved candidate hypotheses, evaluate the candidate RF interrogation configuration according to a predicted distinction between the respective RF responses, and select the candidate RF interrogation configuration when a distinction criterion is satisfied, the candidate RF interrogation configuration comprising one or more of a participating radio-frequency perception node, a
- 25 transmitter-receiver relationship, a spectral layer, a transmit waveform, a beamforming configuration, a polarization configuration, a relative
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- transmission phase or amplitude, a transmission timing relationship, an antenna-element subset and an observation duration; wherein the dynamic perception controller is further configured to cause acquisition of additional RF observations using the selected RF interrogation configuration, associate the additional RF observations with an observation relationship in the RF Spatial World Model, compare the additional RF observations with the predicted RF responses, and cause the spatial reconstruction engine to update the candidate electromagnetic propagation hypotheses and candidate physical spatial hypotheses using the comparison.
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- 10 **4.** The system as claimed in claim 1, further comprising an interface subsystem configured to provide, from the RF Spatial World Model, a machine-readable physical-world state to a plurality of external software systems, wherein at least two of the external software systems use the provided physical-world state for different application-specific processing while the underlying radio-frequency perception infrastructure and RF Spatial World Model are shared.
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- 5.** A method for retrospective reconstruction of a physical environment, comprising: acquiring phase-associated radio-frequency observations from a plurality of spatially distributed radio-frequency perception nodes; storing the radio-frequency observations together with node spatial metadata, synchronization metadata, calibration metadata, transmitter and receiver provenance, and configuration information identifying active transmitter-receiver relationships; storing or otherwise retaining a reconstructability state comprising at least one candidate electromagnetic propagation hypothesis or candidate physical spatial hypothesis and associated confidence and/or
- 20
- uncertainty information generated from the radio-frequency observations; retrieving a selected temporal portion of the stored radio-frequency observations together with the associated reconstructability state and spatial, synchronization, calibration, provenance and configuration information; and reconstructing, from the retrieved information, a historical three-dimensional spatial state of at least a portion of the physical environment, including
- 25
- reconstructing one or more alternative physical spatial hypotheses where the
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retrieved observations remain ambiguous, wherein the historical three-dimensional spatial state is reconstructable for a computationally selected viewpoint within a reconstructable spatial region without requiring a physical radio-frequency perception node to have occupied the selected viewpoint at the historical time.

6. The system as claimed in claim 1, wherein at least one RF perception node comprises an active transmission aperture and a receiving aperture separated by an electromagnetic isolation or decoupling structure, wherein transmit-path and receive-path calibration information is separately represented and applied to compensate node-specific coupling, delay, phase and/or amplitude effects, and wherein calibrated complex-valued observations are supplied to the spatial reconstruction engine.
7. The system as claimed in claim 6, wherein an antenna geometry of at least one of the active transmission aperture and the receiving aperture, and/or a relative geometry between the apertures, is selected or configured according to a spatial reconstruction requirement comprising at least one candidate physical spatial hypothesis, candidate electromagnetic propagation hypothesis, desired spatial resolution or required observation diversity, such that resulting transmitter-receiver relationships provide selected spatial baselines for the reconstruction operation.
8. The system as claimed in claim 7, wherein the selected antenna geometry defines an observation operator having spatial response vectors associated with candidate spatial or propagation states, and wherein the geometry is selected or configured according to a criterion comprising response separation, correlation, rank, conditioning or another measure of distinguishability between at least two candidate states.
9. The system as claimed in claim 1, wherein the plurality of RF perception nodes support active, passive, bistatic, multistatic, cooperative and/or hybrid sensing modes, and wherein observations acquired before and after a change in sensing mode remain spatially registered and are incorporated into a common persistent RF Spatial World Model and associated reconstruction state.

10. The system as claimed in claim 1, wherein the plurality of RF perception nodes operate across multiple spectral layers having different propagation and spatial-resolution characteristics, and wherein observations from the spectral layers are spatially registered to a common physical region and incorporated into a common reconstruction state.
11. The system as claimed in claim 10, wherein a lower-frequency reconstruction produces one or more candidate spatial states, occupancy constraints, boundary constraints and/or candidate electromagnetic propagation hypotheses for a physical region, and wherein a higher-frequency reconstruction is restricted, initialized and/or regularized using a computationally derived admissible solution space based on the lower-frequency reconstruction, such that candidate higher-frequency states inconsistent with the lower-frequency spatial or propagation constraints are excluded, down-weighted or otherwise reduced in confidence.
12. The system as claimed in claim 1, wherein adjacent RF perception nodes provide overlapping effective RF perception volumes for a common physical region, and wherein a coherence-qualified subset of the overlapping nodes is selected for a selected phase-sensitive reconstruction operation while observations from other overlapping nodes are spatially registered and fused as non-coherent or semi-coherent spatial evidence for the same region.
13. The system as claimed in claim 12, wherein the overlapping perception volumes are configured to provide at least two geometrically distinct transmitter-receiver observation relationships for at least part of the common physical region, thereby providing spatially redundant evidence for evaluating candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses.
14. The system as claimed in claim 12, wherein the coherence-qualified subset is defined for the selected reconstruction operation according to a relative timing, frequency and/or phase-error tolerance, and wherein a node may contribute to the common RF Spatial World Model through spatially registered non-coherent

observations when its coherence state does not satisfy the tolerance for the selected phase-sensitive operation.

15. The system as claimed in claim 14, wherein a plurality of coherence subdomains are spatially fused to generate a common RF Spatial World Model.
- 5 16. The system as claimed in claim 2, wherein the transmitter-associated signal characteristics comprise at least one of a carrier-frequency assignment, carrier-frequency offset, phase sequence, code sequence, waveform characteristic, modulation characteristic, pulse timing, pilot structure, chirp characteristic, spreading sequence, subcarrier assignment, waveform identifier and
10 frequency-hopping sequence.
17. The system as claimed in claim 2, wherein transmitter-associated signal characteristics are uniquely or sufficiently distinguishable within a defined RF perception domain and are reusable in a geographically separated RF perception domain.
- 15 18. The system as claimed in claim 1, wherein a spatial processing subsystem generates a Spatiotemporal RF Observation Graph comprising transmitter identity, receiver identity, temporal information, frequency information, phase information and spatial information, and wherein graph relationships associate observations according to at least one of common transmitter, common
20 receiver, temporal proximity, common spatial region, cross-frequency correspondence and candidate propagation relationship.
19. The system as claimed in claim 1, wherein received electromagnetic observations are processed as candidate direct, reflected, scattered, refracted and/or diffracted propagation components, and wherein a candidate
25 propagation component is retained as reconstruction evidence when consistent with at least one transmitter-receiver relationship, spatial geometry, temporal relationship or physical constraint.
20. The system as claimed in claim 1, wherein the spatial reconstruction engine maintains, for at least a portion of the physical environment, a plurality of
30 candidate physical spatial hypotheses, each candidate physical spatial hypothesis being associated with one or more candidate electromagnetic

5 propagation hypotheses and an associated confidence, probability and/or uncertainty value, and wherein the spatial reconstruction engine evaluates subsequently acquired RF observations against the candidate electromagnetic propagation hypotheses and uses resulting consistency or residual information to increase, decrease, retain, split, merge and/or eliminate at least one candidate physical spatial hypothesis.

10 **21.** The system as claimed in claim 20, wherein the spatial reconstruction engine comprises a learned inverse model and an electromagnetic forward model, the learned inverse model being configured to generate or update candidate physical spatial hypotheses from measured RF observations, the electromagnetic forward model being configured to generate predicted RF observations from respective candidate physical spatial hypotheses and associated candidate electromagnetic propagation hypotheses, and wherein a discrepancy between a measured RF observation and a predicted RF
15 observation is used to update the confidence, probability, uncertainty and/or structural state of at least one candidate hypothesis during reconstruction.

20 **22.** The system as claimed in claim 21, wherein the learned inverse model comprises a physics-informed neural network, neural field, graph neural network, convolutional neural network, transformer network, occupancy network, probabilistic model or combination thereof, and wherein the learned inverse model is used to generate candidate physical spatial states rather than being required to directly output a final deterministic RF Spatial World Model state.

25 **23.** The system as claimed in claim 22, wherein the spatial reconstruction engine applies one or more physical consistency constraints to the candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses, the constraints comprising electromagnetic propagation consistency, spatial geometry, material properties, temporal continuity, kinematic feasibility, cross-node consistency and/or cross-frequency consistency, and wherein a constraint
30 violation modifies a confidence, probability, admissibility or structural state associated with at least one candidate hypothesis.

24. The system as claimed in claim 1, wherein the RF Spatial World Model maintains, for at least a portion of the physical environment, both a reconstructed spatial state and a hypothesis-dependent reconstruction state, the hypothesis-dependent reconstruction state retaining at least two competing candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses together with associated observation provenance, confidence and/or uncertainty, and wherein the world model distinguishes between spatial state sufficiently resolved for downstream use and spatial state retained as provisional or hypothesis-dependent pending additional RF evidence.
25. The system as claimed in claim 3, wherein the dynamic perception controller defines a plurality of candidate effective distributed sensing apertures, each candidate effective distributed sensing aperture comprising a selected subset of geographically distributed radio-frequency perception nodes, antenna elements, transmitters, receivers and/or transmitter-receiver relationships; determines, for each candidate effective distributed sensing aperture, an observation operator corresponding to the selected subset and determines or estimates respective radio-frequency responses that would result from applying the candidate effective distributed sensing aperture under at least two competing candidate physical spatial hypotheses and/or associated candidate electromagnetic propagation hypotheses; compares the respective predicted radio-frequency responses under the competing candidate hypotheses to determine a predicted distinguishability; and selects one of the candidate effective distributed sensing apertures based at least in part on the predicted distinguishability.
26. The system as claimed in claim 3, wherein a plurality of candidate RF interrogation configurations differ by respective spectral layers or frequency ranges, and wherein, for at least two competing candidate physical spatial hypotheses and/or associated candidate electromagnetic propagation hypotheses, the dynamic perception controller determines or estimates respective radio-frequency responses that would result from applying the

5 respective candidate spectral layers or frequency ranges under the respective
candidate hypotheses, compares the respective predicted radio-frequency
responses, and selects a spectral layer or frequency range based at least in part
on a predicted distinguishability between the respective radio-frequency
responses.

10 27. The system as claimed in claim 3, wherein the candidate RF interrogation
configuration comprises a controlled change in an electromagnetic sensing,
source or illumination condition between passive, active, bistatic, multistatic
and/or hybrid operation, and wherein the dynamic perception controller
determines or estimates, under at least two competing candidate physical
spatial hypotheses and/or associated candidate electromagnetic propagation
hypotheses, respective predicted radio-frequency response transitions resulting
from the controlled change, compares the respective predicted radio-frequency
response transitions, and selects the controlled change based at least in part on
15 a predicted distinguishability between the respective response transitions,
wherein observations acquired before and after the controlled change remain
associated with the same reconstruction state for updating the candidate
hypotheses.

20 28. The system as claimed in claim 3, wherein the dynamic perception controller
determines, from at least two competing candidate physical spatial hypotheses
and/or associated candidate electromagnetic propagation hypotheses, an
observation-fidelity requirement identifying one or more radio-frequency
information characteristics required to distinguish the competing candidate
hypotheses, and selectively controls acquisition, retention, preprocessing
25 and/or transmission of an observation according to the observation-fidelity
requirement, the radio-frequency information characteristics comprising one or
more of complex-valued in-phase and quadrature information, phase precision,
temporal sampling resolution, frequency resolution, bandwidth, Doppler
resolution, polarization information, propagation-component information,
30 dynamic range and observation duration, and wherein a computational

processing location for the observation is selected according to the observation-fidelity requirement and one or more resource constraints.

- 5 **29.** The system as claimed in claim 3, wherein the infrastructure maintains RF perception experience records, each experience record associating a prior reconstruction condition comprising one or more candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses with a corresponding RF interrogation configuration, hypothesis-specific predicted radio-frequency response, acquired radio-frequency response and reconstruction outcome indicating an effect of the RF interrogation configuration on resolution of the candidate hypotheses; and wherein, upon
10 occurrence of a later reconstruction condition having a selected similarity to the prior reconstruction condition, the dynamic perception controller retrieves the experience record and uses the corresponding RF interrogation configuration to propose or evaluate a subsequent RF interrogation
15 configuration for the later reconstruction condition, subject to validation against the candidate hypotheses and predicted radio-frequency responses of the later reconstruction condition.
- 20 **30.** The method as claimed in claim 5, wherein the stored radio-frequency observations comprise complex-valued I/Q observations, phase information, timestamps, node spatial coordinates, synchronization state, calibration state and transmitter and receiver identifiers, and wherein the stored reconstructability state identifies at least one active transmitter-receiver relationship and at least one candidate electromagnetic propagation or physical-state hypothesis associated with the observation interval.
- 25 **31.** The method as claimed in claim 5, wherein the historical three-dimensional spatial state is reconstructed from a computationally selected viewpoint within a reconstructable spatial region and from the stored reconstructability state associated with the historical observation interval.
- 30 **32.** The method as claimed in claim 5, further comprising cryptographically authenticating the stored radio-frequency observations using a hardware security subsystem.

- 5 **33.** The system as claimed in claim 1, wherein the RF Spatial World Model is maintained at a plurality of spatial resolutions and comprises at least one of a voxel representation, occupancy representation, point representation, mesh, scene graph, neural field, spatial tensor, graph structure or combination thereof, with higher-resolution reconstruction selectively generated for a subregion having elevated uncertainty, increased object density, detected motion or a specified reconstruction requirement.
- 10 **34.** The system as claimed in claim 4, wherein the interface subsystem provides at least one of object geometry, occupancy, position, velocity, material or scattering characteristics, spatial relationships, confidence, uncertainty and temporal state to the plurality of external software systems, and wherein the provided RF Spatial World Model is available as a perception service to the external software systems without requiring conventional optical image data to be supplied as the primary physical-world representation.
- 15 **35.** The system as claimed in claim 34, wherein at least two independent downstream software systems consume the same RF Spatial World Model while executing different application-specific analysis, control, optimization, security, logistics, industrial, mobility or analytics functions.
- 20 **36.** The system as claimed in claim 3, wherein the selected RF interrogation configuration and corresponding hypothesis-specific predicted radio-frequency responses are retained with an observation relationship in the RF Spatial World Model, and wherein, after acquisition of an RF observation using the selected RF interrogation configuration, the dynamic perception controller compares the acquired RF observation with the retained predicted radio-frequency responses, updates confidence, probability, uncertainty and/or admissibility associated with at least one candidate electromagnetic propagation hypothesis and/or candidate physical spatial hypothesis, and stores the resulting reconstruction state and observation provenance in the RF Spatial World Model.
- 25 **37.** The system as claimed in claim 1, wherein sensing-resource selection is performed according to an uncertainty representation associated with the RF
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Spatial World Model and at least one candidate electromagnetic propagation hypothesis associated with the uncertainty representation, and wherein at least one candidate sensing configuration is evaluated according to an expected ability to distinguish the candidate electromagnetic propagation hypothesis and/or candidate physical spatial hypothesis from another candidate hypothesis, subject to one or more communication, energy, spectrum, synchronization and computation constraints.

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38. The system as claimed in claim 1, wherein at least one RF perception node operates in a passive mode using ambient electromagnetic signals and at least one other RF perception node operates in an active transmission mode, and wherein observations acquired in the passive and active modes are spatially registered and associated with a common reconstruction state for the physical region.

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39. The system as claimed in claim 1, wherein an environmental or physical-world state obtained from a lower-frequency spectral layer produces one or more spatial, occupancy, boundary and/or candidate electromagnetic propagation constraints that define or restrict an admissible higher-frequency hypothesis space comprising at least two surviving candidate physical spatial hypotheses and/or candidate electromagnetic propagation hypotheses, and wherein a dynamic perception controller determines or estimates respective radio-frequency responses that would result from applying a candidate higher-frequency RF interrogation configuration under at least two of the surviving candidate hypotheses, compares the respective predicted radio-frequency responses, and selects or evaluates the candidate higher-frequency RF interrogation configuration based at least in part on a predicted distinguishability between the respective radio-frequency responses, with a resulting higher-frequency reconstruction being spatially registered with the lower-frequency state in the RF Spatial World Model.

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40. The system as claimed in claim 1, wherein the inter-node synchronization and calibration subsystem maintains a time-varying hardware calibration state for at least one radio-frequency perception node; obtains at least one of internal

calibration observations, reciprocal inter-node calibration observations and operating-condition measurements; compares subsequent radio-frequency observations with persistent spatial entities represented in the RF Spatial World Model; generates competing physical-environment-change, electromagnetic-propagation-change and hardware-calibration-drift hypotheses; and updates the hardware calibration state and/or at least one candidate physical spatial hypothesis according to physical consistency, cross-node consistency, calibration residual, synchronization residual and associated uncertainty.

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Date: KRISHNA
PRASAD
15 **Signature of Applicant:** SIDHEESHWARAN
Krishna Prasad Sidheeshwaran
Digitally signed by
KRISHNA PRASAD
SIDHEESHWARAN
Date: 2026.09.26
23:10:26 +05'30'

ABSTRACT OF THE INVENTION

A RADIO-FREQUENCY PERCEPTION-AS-INFRASTRUCTURE SYSTEM AND METHOD FOR DISTRIBUTED THREE-DIMENSIONAL SPATIAL WORLD MODEL GENERATION

The invention relates to distributed radio-frequency sensing and spatial computing for generating a persistent machine-readable three-dimensional representation of a physical environment. A plurality of geographically distributed RF perception nodes acquire active, passive and/or cooperative electromagnetic observations using multiple transmitter-receiver relationships, spectral layers and sensing configurations. Inter-node synchronization and calibration establish temporal, frequency and/or phase relationships, while transmitter identification, spatial registration and a Spatiotemporal RF Observation Graph associate observations with candidate electromagnetic propagation and physical spatial hypotheses. A spatial reconstruction engine evaluates the hypotheses using electromagnetic forward models, learned inverse models and physical consistency constraints, and maintains confidence and uncertainty within an RF Spatial World Model. The uncertainty state controls subsequent sensing and computational resources, enabling adaptive distributed reconstruction. The resulting world model is provided as a perception service to autonomous, robotic, industrial, security and analytical systems.

FIG. 7